

Netra

A complete, in-depth explanation of the Integrated CCTV Monitoring, Video Analytics and Real-Time Alert Platform — what every part does, how it works internally, and why it was built that way.

Prepared for	okDriver Smart Dashcams Private Limited
Candidate	Aditya Kaushik · Full Stack Developer (Intern)
Source code	github.com/AdityaKaushik11/netra
Companion	docs/Netra-Architecture.pdf (24-page architecture summary)
Document date	2 October 2026 · version 1.0

Every value in this handbook (thresholds, limits, ports, table and column names, endpoints) is taken from the source code and configuration in the repository. Screenshots are of the running system.

Contents

Part I — The product

- 1 What Netra is and the problem it solves
- 2 A guided tour of the console
- 3 Running the project

Part II — Architecture

- 4 System architecture
- 5 Repository walkthrough
- 6 Runtime: containers, networks, start-up

Part III — Backend in depth

- 7 Configuration
- 8 Data model, column by column
- 9 Identity, roles and sessions
- 10 Camera registry and source adapters
- 11 Camera health monitoring
- 12 Video: live, WebRTC, recording
- 13 The ingestion pipeline
- 14 Watchlist matching algorithms
- 15 Alerts
- 16 Search and movement tracing
- 17 Real-time delivery
- 18 Statistics, metrics and audit
- 19 Data retention
- 20 API reference

Part IV — Frontend in depth

- 21 Frontend architecture
- 22 Key components
- 23 Pages

Part V — Simulators and infrastructure

- 24 The simulated world
- 25 Video gateway, proxy and Compose

Part VI — Quality, security, operations

- 26 Security model
- 27 Testing and CI
- 28 Deployment
- 29 Scaling to 80,000 cameras
- 30 Troubleshooting
- 31 Limitations and roadmap
- 32 Glossary

PART I

The product

What the platform is for, what an operator sees and does, and how to run it in one command.

1 What Netra is and the problem it solves

A city's CCTV estate is heterogeneous: traffic-police ANPR cameras, RTO checkpoints, PTZ cameras behind ONVIF, an older video-management system that only publishes HLS, a vendor NVR that exposes nothing but a REST API, and — for okDriver — dashcams on moving vehicles. Each speaks a different protocol, has its own console, and none of them knows which vehicles the police are looking for.

Netra puts all of them behind one platform. It does five things:

- 1 **Registers and monitors every camera** — whatever its protocol — and knows at all times whether it is Online, Degraded or Offline and why.
- 2 **Shows every feed in one console** — live (low-latency HLS or sub-second WebRTC) and recorded.
- 3 **Receives AI inference results** (number-plate reads, faces, objects, dashcam driver-safety events), validates them, removes duplicates and stores them.
- 4 **Correlates every detection with a watchlist** of stolen or blacklisted vehicles and wanted or missing persons, and raises an alert on every operator's screen in under a second when something matches.
- 5 **Reconstructs movement** — where a vehicle was seen, in what order, how fast it must have travelled — on a map.

Who uses it

Role	Typical person	What they can do
ADMIN	Platform / control-room administrator	Everything: onboard and edit cameras, manage users and API keys, read the audit log, plus all operator actions
OPERATOR	Control-room operator on shift	Acknowledge and resolve alerts, manage the watchlist, probe cameras, export reports, plus all viewer actions
VIEWER	Analyst, supervisor, auditor	Read-only access to the dashboard, cameras, video, detections, alerts, trace and watchlist
API client	AI engine, edge gateway, dashcam, partner VMS	Machine identity with explicit scopes: post events, send heartbeats, onboard cameras

A day in the life — the story the system is built around

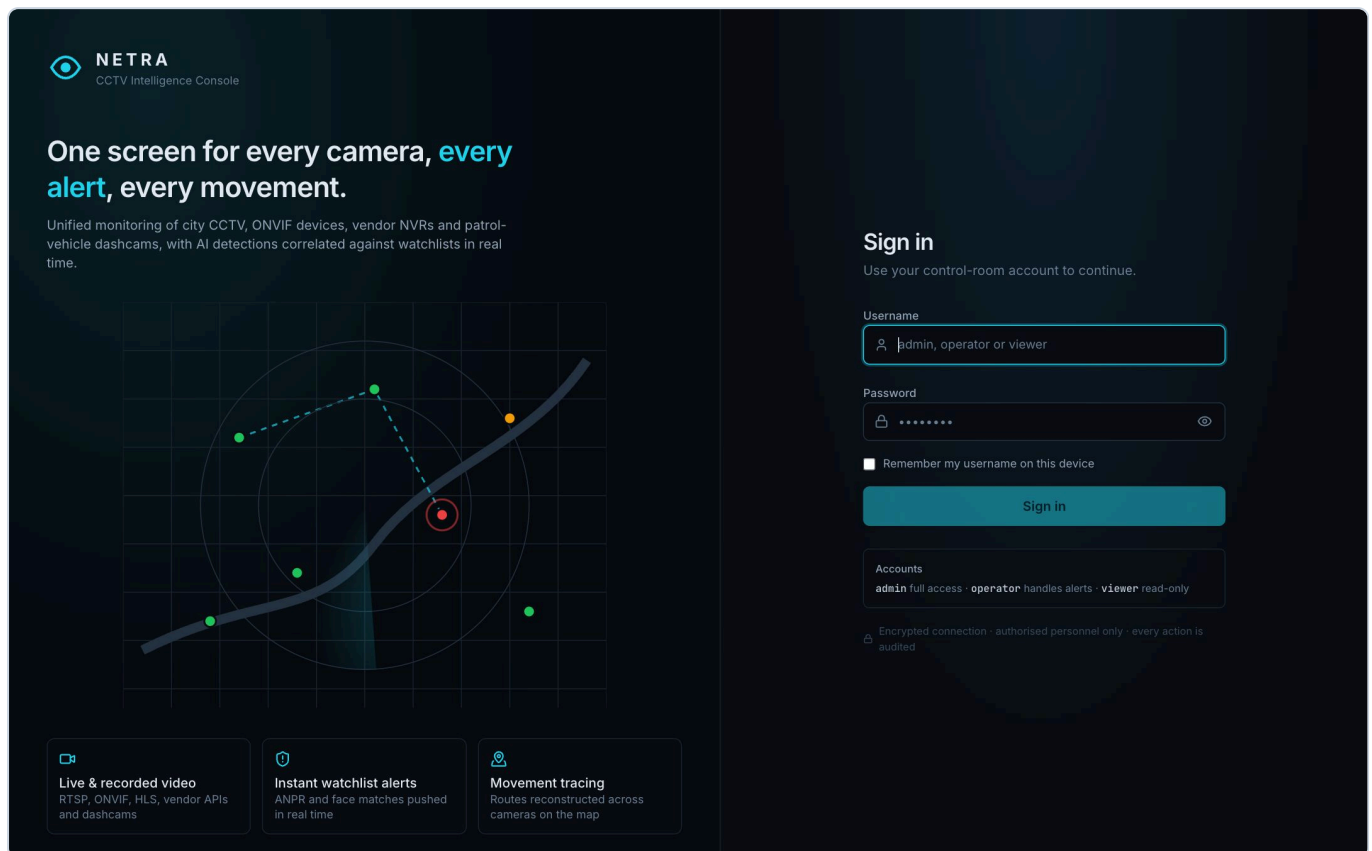
A white Swift Dzire, GJ01XX0001, is reported stolen; an operator adds it to the watchlist as CRITICAL. At 10:02 an ANPR camera at Paldi Cross Roads reads the plate. Within a second every operator console beeps three times, a red toast shows the plate, camera, time and confidence, and the Paldi marker on the map starts pulsing. The operator opens the alert: the watchlist record (make, colour, FIR number), the detection in plain language, the camera's live feed, a 40-second recorded clip of the moment, and a map showing the operator's position, the car's position, the distance and the drive time. They acknowledge it ("PCR 12 dispatched"). At 10:18 the car passes the Subhash Bridge RTO checkpoint, at 10:41 Vaishnodevi Circle — each sighting updates the trace, which now draws the route across the city. When the vehicle is intercepted the operator resolves the alert with a note; every step is in the audit log.

Observed in operation: while this handbook was written the stack had been running continuously for six days on a laptop, had processed more than 90,000 simulated detections, and recovered automatically each time the laptop slept and woke.

2 A guided tour of the console

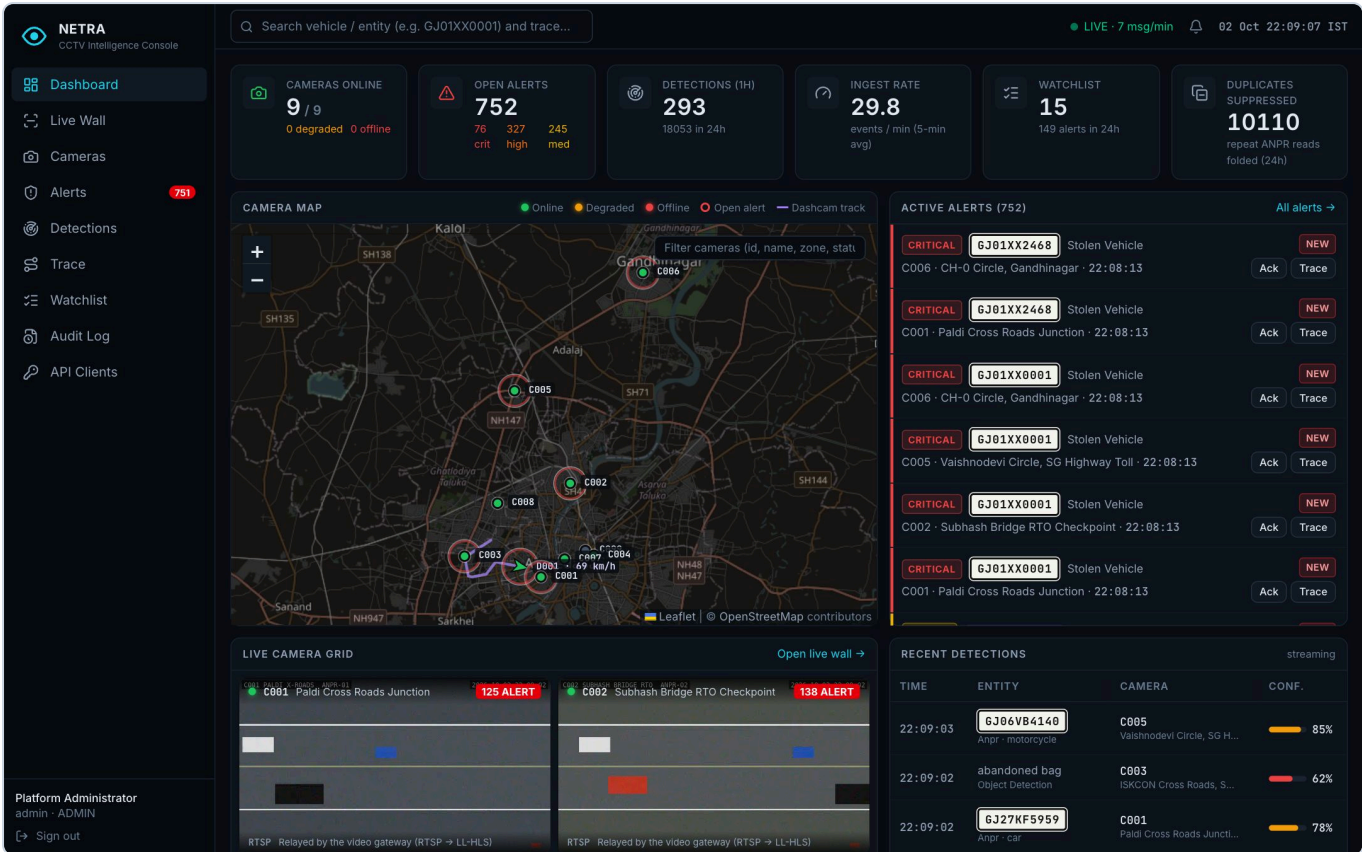
Every screen below is a screenshot of the running system with simulated traffic.

2.1 Sign in



The sign-in page. Username + password; show/hide password, Caps-Lock warning, "remember my username", and a clear message if someone types an e-mail address. Five failed attempts lock the account for 15 minutes.

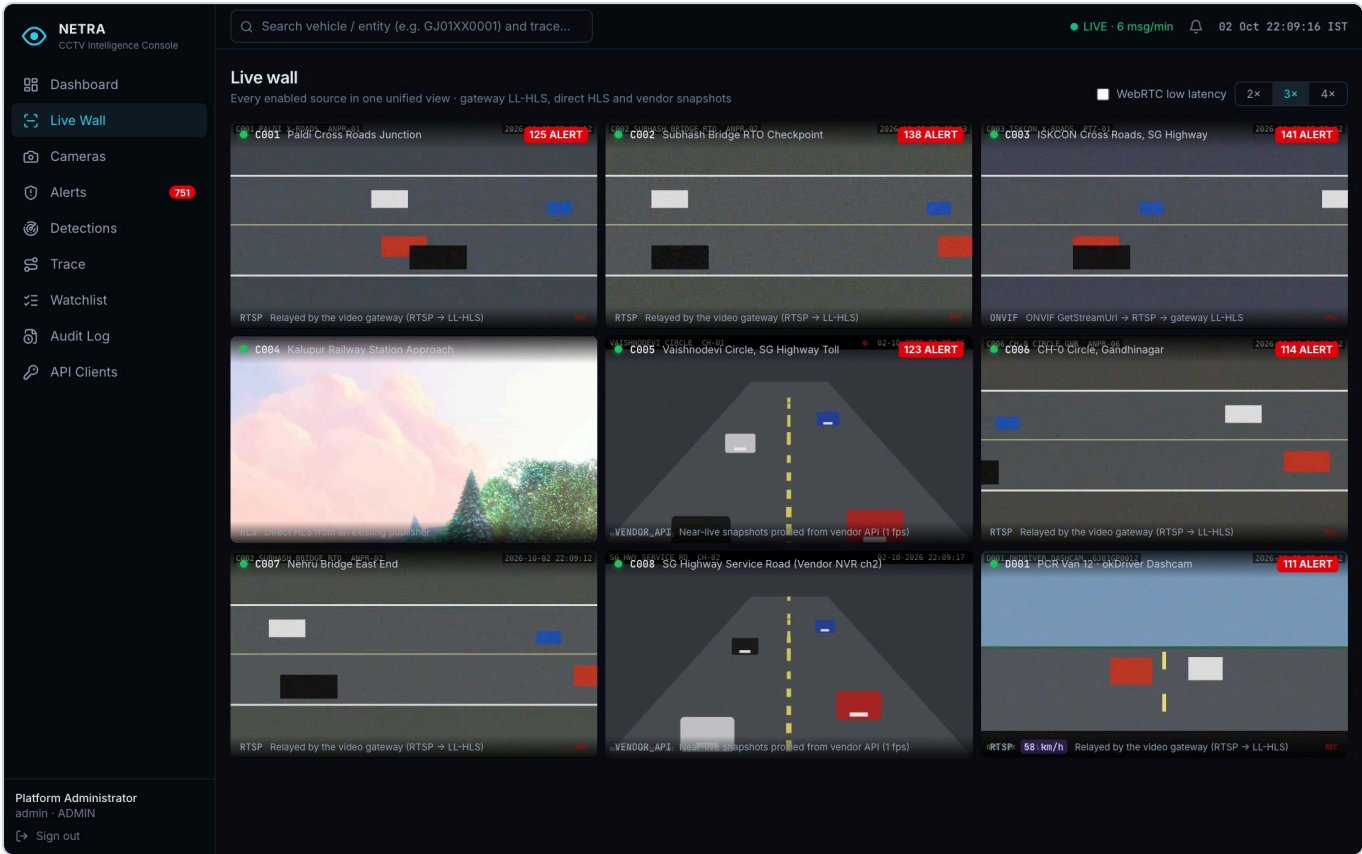
2.2 Dashboard — the operator's single screen



Dashboard.

Area	What it shows and how it stays live
KPI strip	Cameras online / total with degraded and offline counts; open alerts split by severity; detections in the last hour and 24 hours; ingest rate (events per minute, 5-minute average); watchlist size and alerts in 24 h; repeat ANPR reads suppressed in 24 h. Re-fetched whenever a real-time message arrives (throttled to once per 1.2 s) and every 30 s.
Camera map	Every camera as a status-coloured dot (green / amber / red / grey); a pulsing red ring when the camera has an open alert; the dashcam as an arrow pointing along its heading with its speed, plus its purple GPS trail. A filter box narrows the map, the grid and the health table. Clicking a camera opens a mini live view.
Active alerts	Newest open alerts with severity colour bar, plate or person reference, category, camera and time; one-click Acknowledge and Trace.
Live camera grid	Four live feeds (fixed cameras plus the dashcam).
Recent detections	Streaming table of detections; a new row flashes as it arrives over the WebSocket; watchlist hits carry a warning icon; repeat reads show "x n"; dashcam ADAS events show their description.
Trend & health	Detections per hour for 24 h (hover for watchlist hits); a service strip (API, database, Redis, gateway, ingest backlog, streams ready, viewers) and every camera's status, reason and last-seen time.

2.3 Live wall



Live wall: all nine cameras and all five source types in one view — RTSP via the gateway (C001, C002, C006, C007), ONVIF (C003), an existing HLS publisher (C004), vendor-NVR snapshots (C005, C008) and the dashcam (D001, with live speed). The 2x / 3x / 4x buttons change the grid; "WebRTC low latency" switches gateway feeds to sub-second WebRTC.

2.4 Camera registry and camera detail

NETRA
CCTV Intelligence Console

Dashboard

Live Wall

Cameras

Alerts

Detections

Trace

Watchlist

Audit Log

API Clients

Platform Administrator
admin · ADMIN

Sign out

Q Search vehicle / entity (e.g. GJ01XX0001) and trace...

LIVE · 2 msg/min

02 Oct 22:09:23 IST

Camera registry

Heterogeneous sources (RTSP, ONVIF, HLS, vendor API) onboarded into one registry

ONVIF discovery

+ Add camera

10 CAMERAS

Search id, name, zone...

All status

All protocols

All departments

All zones

ID	NAME / LOCATION	DEPT · ZONE	SOURCE	STATUS	LAST SEEN	STORAGE
C001	Paldi Cross Roads Junction 23.0120, 72.5626 · Anpr	AHM-TRF Ahmedabad West	RTSP active probe · seed	ONLINE 125 stream ready on gateway	4 seconds ago	HOT · 30d recording
C002	Subhash Bridge RTO Checkpoint 23.0746, 72.5838 · Anpr	RTO-GJ Ahmedabad North	RTSP active probe · seed	ONLINE 138 stream ready on gateway	4 seconds ago	HOT · 30d recording
C003	ISKCON Cross Roads, SG Highway 23.0258, 72.5073 · Ptz	AHM-CITY Ahmedabad West	ONVIF active probe · seed	ONLINE 141 stream ready on gateway	4 seconds ago	WARM · 30d recording
C004	Kalupur Railway Station Approach 23.0269, 72.6008 · Dose	AHM-CITY Ahmedabad East	HLS active probe · seed	ONLINE manifest OK	4 seconds ago	WARM · 30d live only
C005	Vaishnodevi Circle, SG Highway Toll 23.1365, 72.5436 · BulLet	AHM-TRF Ahmedabad North	VENDOR API active probe · seed	ONLINE 123 vendor API OK	4 seconds ago	HOT · 30d recording
C006	CH-0 Circle, Gandhinagar 23.2156, 72.6369 · Anpr	GNR-POL Gandhinagar	RTSP edge heartbeat · seed	ONLINE 114 heartbeat OK	11 seconds ago	HOT · 30d recording
C007	Nehru Bridge East End 23.0243, 72.5800 · Anpr	AHM-TRF Ahmedabad East	RTSP active probe · api	ONLINE stream ready on gateway	4 seconds ago	HOT · 30d recording
C008	SG Highway Service Road (Vendor NVR ch2) 23.0612, 72.5309 · BulLet	AHM-TRF Ahmedabad North	VENDOR API active probe · api	ONLINE vendor API OK	4 seconds ago	WARM · 15d live only
C099	UI Test Junction 23.0380, 72.5950 · Anpr	— Ahmedabad East	HLS active probe · manual	DISABLED disabled by administrator	6 days ago	HOT · 30d recording
PCR Van 12 · okDriver		AHM-CITY	RTSP	ONLINE 111		HOT · 15d

LOCATIONS

Kalol

Gandhinagar

Adalaj

NH147

Chandiyala Taluka

Sarkhej

Vatva Taluka

Leaflet

OpenStreetMap contributors

Camera registry: search and filter by status, protocol, department and zone; each row shows source type, health mode, onboarding source, status with reason, last seen and storage policy; locations on the right.

C001 Paldi Cross Roads Junction ONLINE

C001 Paldi Cross Roads Junction

2026-12-25 ALERT

RTSP: Relayed by the video gateway (RTSP → LL-HLS)

Low-latency live view (WebRTC, falls back to LL-HLS)

Recorded footage

Last 1 min

5 min ago

15 min ago

Health

stream ready on gateway

Tracks: H264 Readers: 2 Bitrate (kbps): 692

Detections

Probe now

Edit

Disable

AUDIT HISTORY

02 Oct 2026, 22:08:58	STREAM_ACCESSED	admin	—
02 Oct 2026, 11:01:58	CAMERA_STATUS_CHANGED	health-monitor	DEGRADED → ONLINE Reason: stream ready on gateway
02 Oct 2026, 11:01:47	CAMERA_STATUS_CHANGED	health-monitor	ONLINE → DEGRADED Reason: probe failed, retrying (gateway cannot pull stream from source)
02 Oct 2026, 08:21:21	CAMERA_STATUS_CHANGED	health-monitor	DEGRADED → ONLINE Reason: stream ready on gateway
02 Oct 2026, 08:21:10	CAMERA_STATUS_CHANGED	health-monitor	ONLINE → DEGRADED Reason: probe failed, retrying (gateway cannot pull stream from source)

Department

Ahmedabad Traffic Police

Zone / address

Ahmedabad West · Paldi Char Rasta, Ahmedabad

Coordinates

23.01203, 72.56261

Type / protocol

ANPR · RTSP

Stream endpoint

rtsp://cams1m:8554/c001

Credentials

viewer · (encrypted)

Gateway path

cam-c001

Vendor / model

Hikvision (simulated) DS-2CD7A26G0

Resolution / FPS

1280×720 @ 15

Storage

HOT · 30 days · s3://netra-hot/ahm-west/C001

Health mode

PROBE

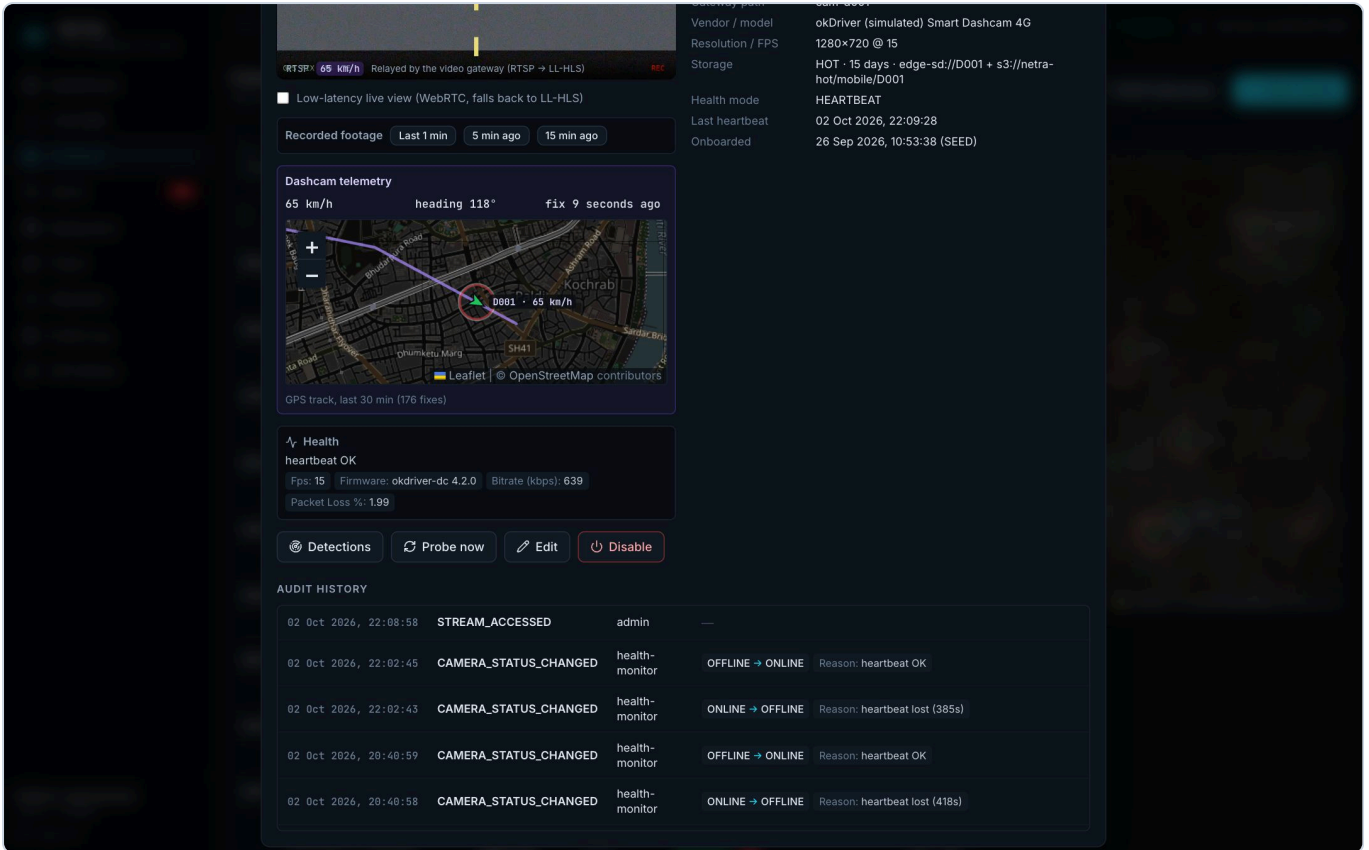
Last heartbeat

02 Oct 2026, 22:09:18

Onboarded

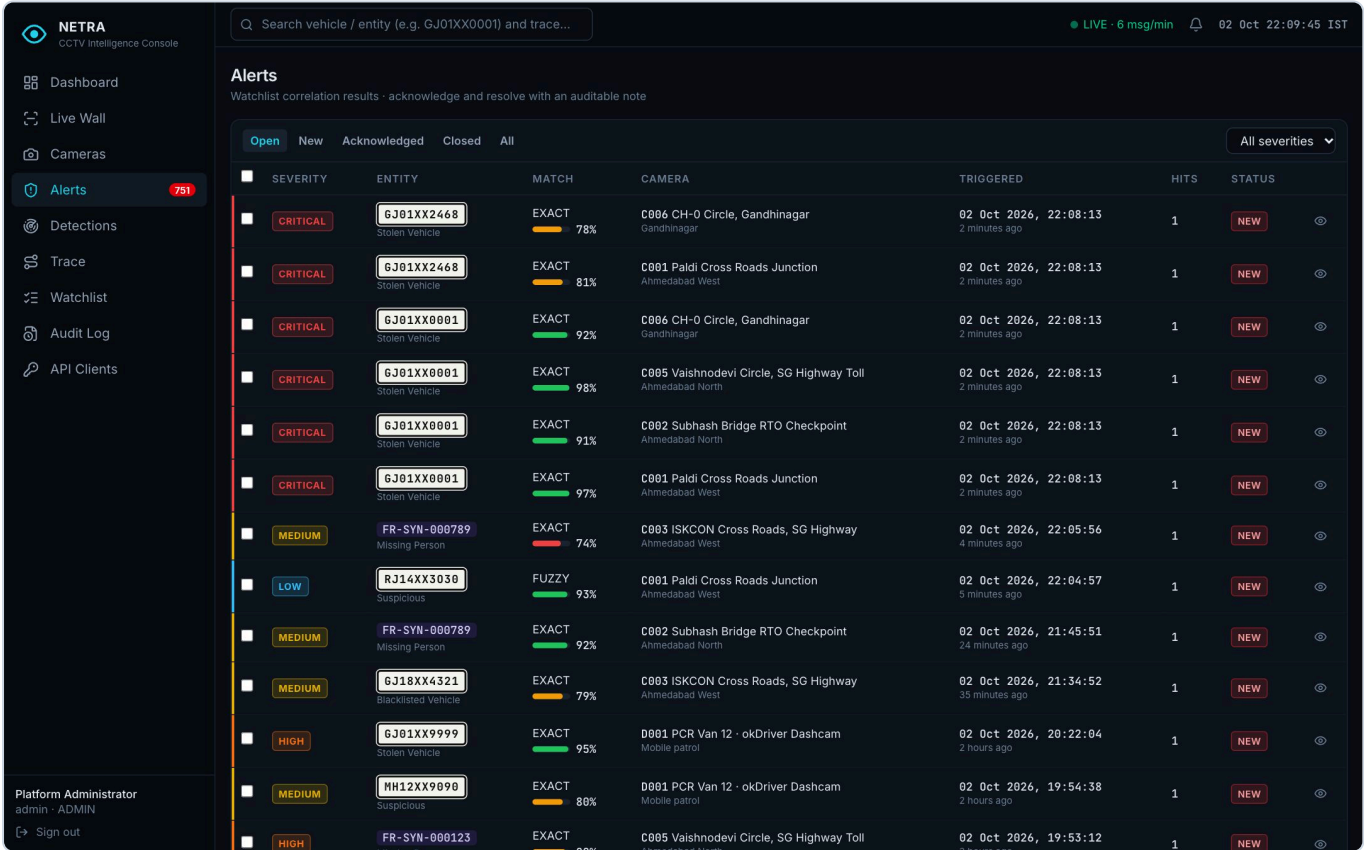
25 Sep 2026, 20:25:29 (SEED)

Camera detail: live view (with optional WebRTC), recorded-footage replay (last 1 / 5 / 15 minutes), health metrics, every metadata field, encrypted-credential indicator, actions (Detections, Probe now, Edit, Disable) and the camera's audit history.



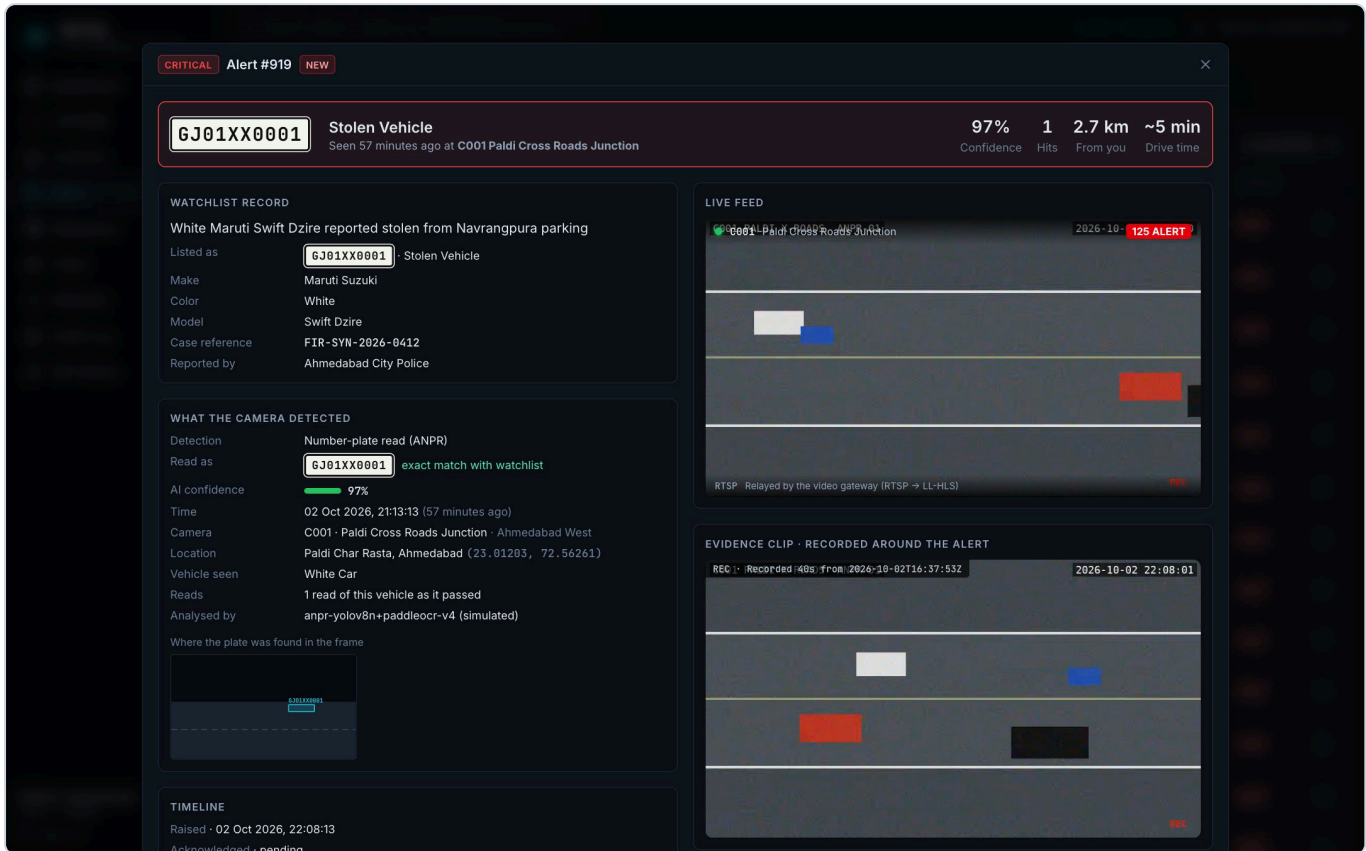
The dashcam D001: live speed, heading and GPS fix age, its 30-minute track on the map, edge health (fps, bitrate, packet loss, firmware) and an audit history that records the moments its heartbeat was lost and restored.

2.5 Alerts

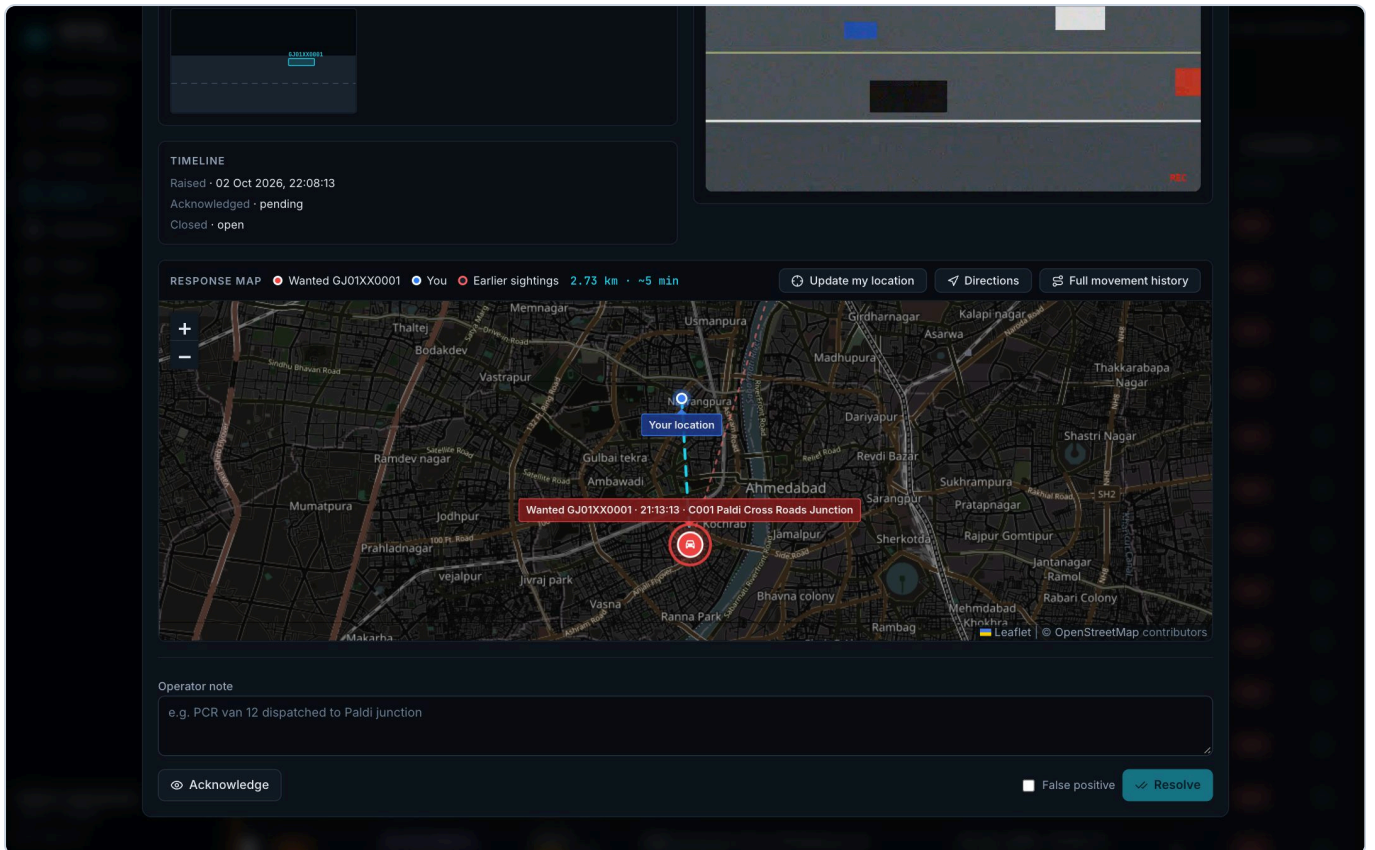


Alert queue with tabs (Open / New / Acknowledged / Closed / All), severity filter, match type and confidence, camera, trigger time and hit count. Tick boxes enable bulk acknowledge / resolve.

The "751" badge is the number of NEW alerts after six days of unattended simulated traffic — nobody was acknowledging them. In a demo the backlog is cleared with the bulk-resolve action (every closure is audited individually).



Alert detail: summary strip (plate, category, last seen, confidence, hits, distance and drive time from the operator), the watchlist record, the detection explained in plain language with where the plate was in the frame, a timeline, the camera's live feed and a recorded 40-second evidence clip.



Response map inside the alert: the operator's own location (from the browser, falling back to the control room), the wanted vehicle's latest sighting, recent earlier sightings, the distance and estimated drive time, and buttons for live directions and the full movement history. Below: operator note, Acknowledge, Resolve, False positive.

2.6 Detections, trace, watchlist, audit, API clients

NETRA

CCTV Intelligence Console

Dashboard

Live Wall

Cameras

Alerts

Detections

Trace

Watchlist

Audit Log

API Clients

Platform Administrator

admin - ADMIN

Sign out

Q Search vehicle / entity (e.g. GJ01XX0001) and trace...

LIVE · 4 msg/min

02 Oct 22:10:02 IST

Detection history

Every validated analytics event, persisted and searchable · repeat reads folded by the dedup window

Export CSV

92032

EVENTS

PLATE / REF

All cameras

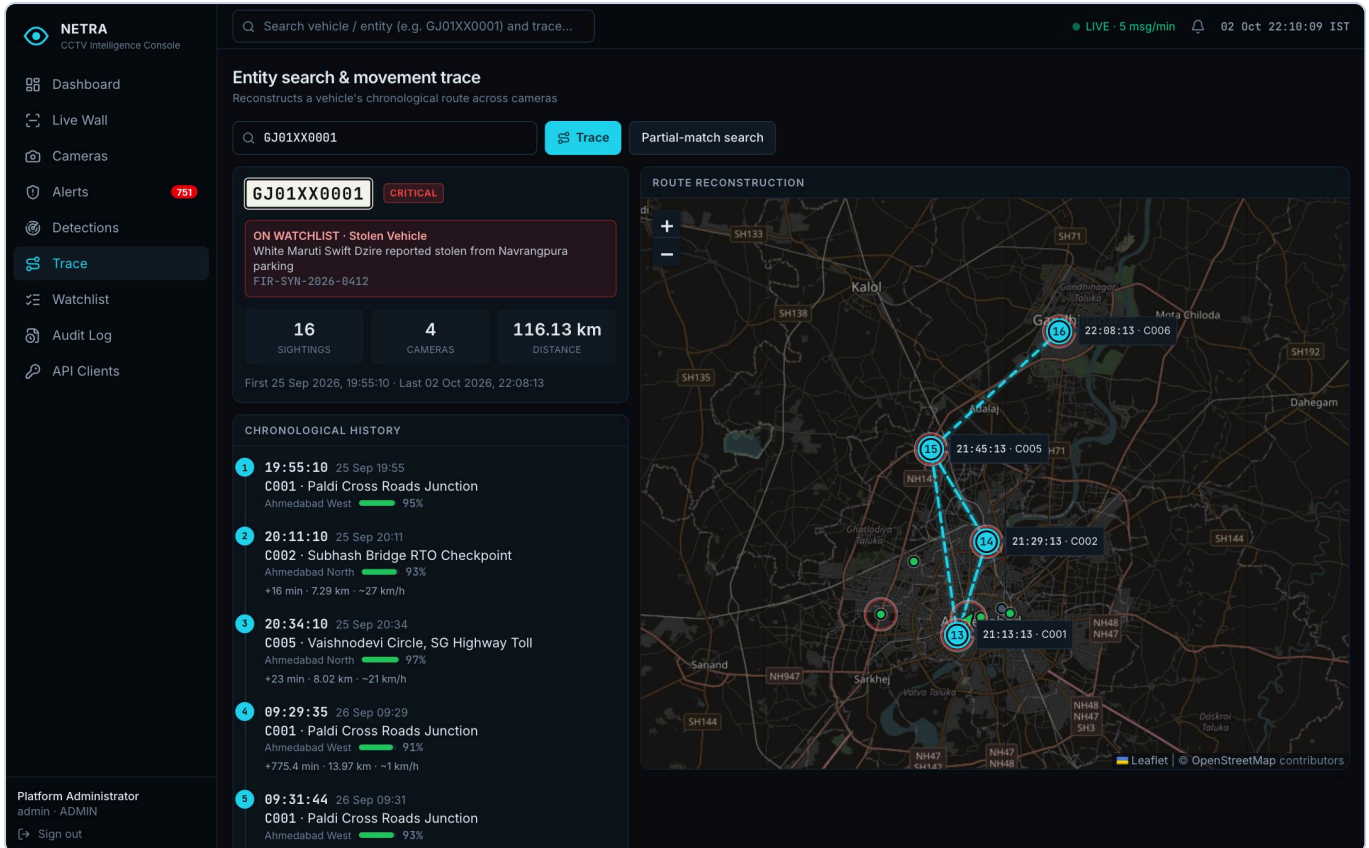
All types

Any

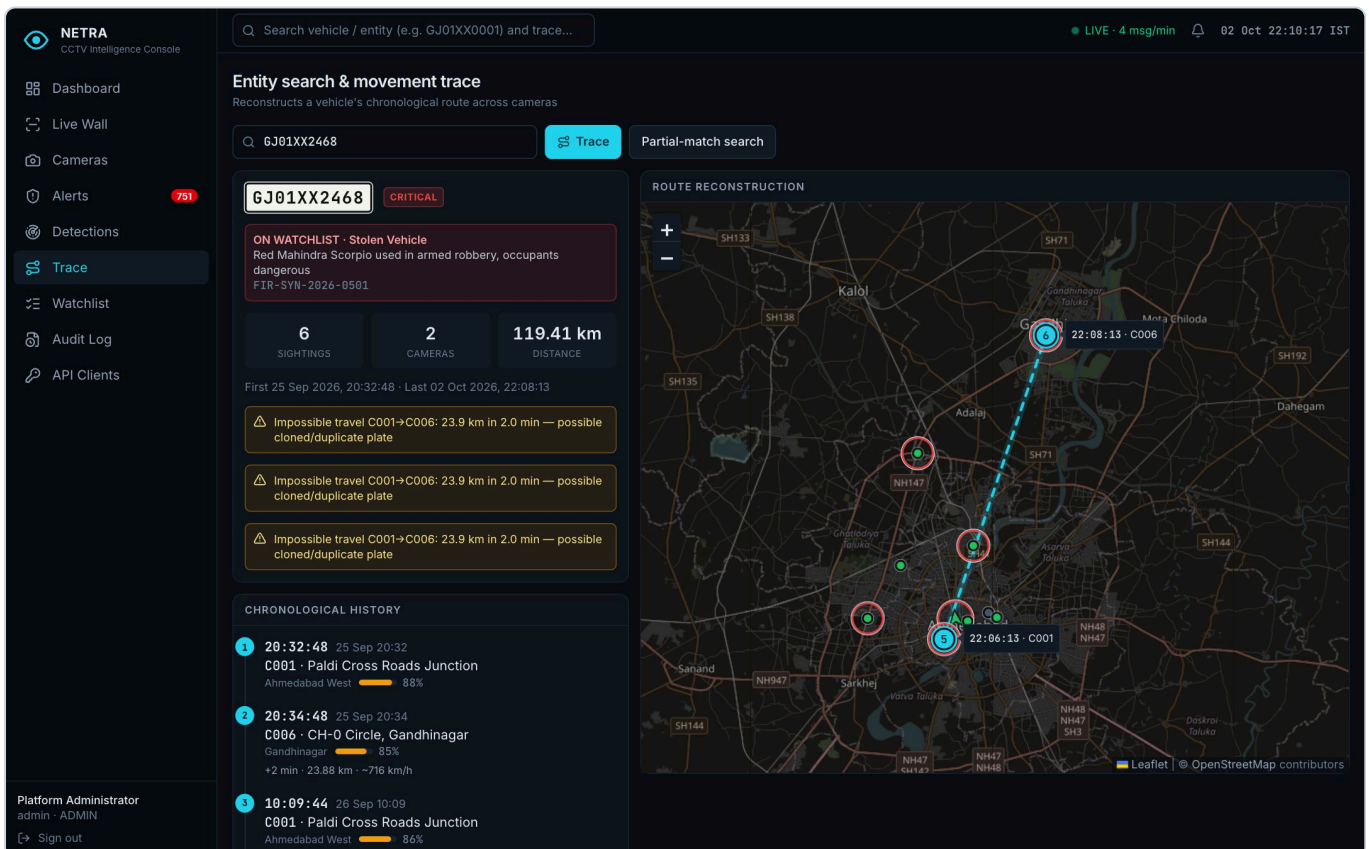
dd/mm/yyyy, --:-- --

DETECTED AT	TYPE	ENTITY	VEHICLE	CAMERA	CONFIDENCE	READS	SOURCE
02 Oct 2026, 22:09:58	Number-plate read (ANPR)	GJ01VS8581	auto_rickshaw	D001 PCR Van 12 · okDriver Dashcam	88%	1	client:analytics-simulator
02 Oct 2026, 22:09:57	Number-plate read (ANPR)	GJ05NP2249	blue car	C006 CH-0 Circle, Gandhinagar	92%	3	stream:analytics-sim
02 Oct 2026, 22:09:56	Number-plate read (ANPR)	MH12ED4458	grey truck	C001 Paldi Cross Roads Junction	92%	3	client:analytics-simulator
02 Oct 2026, 22:09:56	Number-plate read (ANPR)	MH12ED4458	silver motorcycle	C001 Paldi Cross Roads Junction	98%	1	stream:analytics-sim
02 Oct 2026, 22:09:52	Number-plate read (ANPR)	GJ18BD3646	red motorcycle	C001 Paldi Cross Roads Junction	96%	1	stream:analytics-sim
02 Oct 2026, 22:09:50	Number-plate read (ANPR)	GJ01ZJ5427	black bus	C003 ISKCON Cross Roads, SG Highway	93%	2	stream:analytics-sim
02 Oct 2026, 22:09:48	Number-plate read (ANPR)	GJ05MH4480	grey motorcycle	C006 CH-0 Circle, Gandhinagar	96%	1	stream:analytics-sim
02 Oct 2026, 22:09:46	Number-plate read (ANPR)	GJ01WU9571	grey auto_rickshaw	C003 ISKCON Cross Roads, SG Highway	79%	1	client:analytics-simulator
02 Oct 2026, 22:09:43	Number-plate read (ANPR)	GJ01KV5879	black bus	C005 Vaishnodevi Circle, SG Highway Toll	90%	2	stream:analytics-sim
02 Oct 2026, 22:09:43	Number-plate read (ANPR)	GJ01AR2590	motorcycle	D001 PCR Van 12 · okDriver Dashcam	86%	1	client:analytics-simulator
02 Oct 2026, 22:09:40	Number-plate read (ANPR)	GJ38NL6049	truck	D001 PCR Van 12 · okDriver Dashcam	96%	1	client:analytics-simulator
02 Oct 2026, 22:09:39	Face recognition	FR-SYN-599482	—	C005 Vaishnodevi Circle, SG Highway Toll	75%	1	client:analytics-simulator
02 Oct 2026, 22:09:38	Number-plate read (ANPR)	GJ06CD0186	red car	C001 Paldi Cross Roads Junction	84%	2	client:analytics-simulator
02 Oct 2026, 22:09:34	Harsh braking	63 → 35 km/h	—	D001 PCR Van 12 · okDriver Dashcam	87%	1	client:analytics-simulator
02 Oct 2026, 22:09:30	Number-plate read (ANPR)	GJ27ZL0713	white car	C001 Paldi Cross Roads Junction	90%	1	stream:analytics-sim
02 Oct 2026, 22:09:28	Overspeeding	65 km/h in 50 zone	—	D001 PCR Van 12 · okDriver Dashcam	89%	1	client:analytics-simulator

Detection history: every stored event with filters (plate prefix, camera, type, watchlist hits, time range), number of reads per event and the producer. Operators can export the filtered list as CSV.



Trace of GJ01XX0001: watchlist banner, sightings / cameras / distance, and the chronological history with time between stops, leg distance and implied speed; the numbered route is drawn on the map.



Cloned-plate detection: GJ01XX2468 seen at Paldi (C001) and then at Gandhinagar CH-0 (C006), 23.9 km apart, two minutes later — about 716 km/h — so the trace flags "Impossible travel — possible cloned/duplicate plate".

NETRA

CCTV Intelligence Console

Dashboard

Live Wall

Cameras

Alerts751

Detections

Trace

Watchlist

Audit Log

API Clients

Platform Administrator

admin · ADMIN

Sign out

Q Search vehicle / entity (e.g. GJ01XX0001) and trace...

LIVE · 3 msg/min

02 Oct 22:10:22 IST

Watchlist

Synthetic representative records · matched against every detection (exact + OCR-tolerant fuzzy)

+ Add entry

15 ENTRIES

Search...

All categories

Active

IDENTIFIER	CATEGORY	SEVERITY	DESCRIPTION	CASE / AGENCY	ALERTS	LAST SEEN		
FR-SYN-000789 PERSON	Missing Person	MEDIUM	Elderly person with dementia, 74 years (synthetic face-gallery id)	MP-SYN-2026-0081 Gandhinagar Police	52	4 minutes ago		
FR-SYN-000456 PERSON	Wanted Person	CRITICAL	Absconding accused, multiple chain-snatching cases (synthetic face-gallery id)	WNT-SYN-2026-0019 State Crime Branch	67	2 hours ago		
FR-SYN-000123 PERSON	Missing Person	HIGH	Missing child, 11 years, last seen at Kankaria Lake (synthetic face-gallery id)	MP-SYN-2026-0077 Ahmedabad City Police	63	2 hours ago		
GJ01XX9999 VEHICLE	Stolen Vehicle	HIGH	Grey Royal Enfield Classic 350 stolen from Maninagar	FIR-SYN-2026-0477 Ahmedabad City Police	116	2 hours ago		
GJ38XX1111 VEHICLE	Blacklisted Vehicle	MEDIUM	Overloading repeat offender (goods carrier)	RTO-SYN-BVD-0098 RTO Bavia	0	—		
DL3CXX4545 VEHICLE	Suspicious	MEDIUM	Fake number plate suspected (duplicate registration)	INT-SYN-0133 State Crime Branch	0	—		
GJ01XX2468 VEHICLE	Stolen Vehicle	CRITICAL	Red Mahindra Scorpio used in armed robbery, occupants dangerous	FIR-SYN-2026-0501 Ahmedabad Crime Branch	6	2 minutes ago		
GJ06XX8080 VEHICLE	Blacklisted Vehicle	LOW	Insurance lapsed > 1 year	RTO-SYN-VAD-1180 RTO Vadodara	1	8 days ago		
RJ14XX3030 VEHICLE	Suspicious	LOW	Observed repeatedly near ATM kiosks after midnight	INT-SYN-0107 State Crime Branch	93	6 hours ago		
GJ01XX5555 VEHICLE	Stolen Vehicle	HIGH	Blue Hyundai Creta stolen from Bodakdev	FIR-SYN-2026-0455 Ahmedabad City Police	113	2 hours ago		
NH12XX9090 VEHICLE	Suspicious	MEDIUM	Inter-state vehicle linked to chain-snatching pattern	INT-SYN-0091 State Crime Branch	101	2 hours ago		
GJ27XX7788 VEHICLE	Blacklisted Vehicle	HIGH	Fitness certificate revoked; involved in hit-and-run inquiry	RTO-SYN-AHM-7781 RTO Ahmedabad	111	2 hours ago		
GJ18XX4321 VEHICLE	Blacklisted Vehicle	MEDIUM	Commercial vehicle with 14 unpaid e-challans, permit suspended	RTO-SYN-GNR-2291 RTO Gandhinagar	90	35 minutes ago		

Watchlist: vehicles and persons with category, severity, description, case reference, issuing agency, number of alerts raised and when the entity was last seen.

NETRA

CCTV Intelligence Console

Dashboard

Live Wall

Cameras

Alerts751

Detections

Trace

Watchlist

Audit Log

API Clients

Platform Administrator

admin · ADMIN

Sign out

Q Search vehicle / entity (e.g. GJ01XX0001) and trace...

LIVE · 4 msg/min

02 Oct 22:10:27 IST

Audit log

Immutable record of logins, onboarding, configuration changes, stream access and alert handling

3783 ENTRIES

All entities

All actions

TIME	ACTOR	ACTION	ENTITY	DETAILS	IP
02 Oct 2026, 22:09:45	admin USER	RECORDING_ACCESSED	camera:C001	Start: 2026-10-02T16:37:53.135000+00:00 Duration: 40	172.18.0.1
02 Oct 2026, 22:09:07	admin USER	STREAM_ACCESSED	camera:C008	—	172.18.0.1
02 Oct 2026, 22:09:07	admin USER	STREAM_ACCESSED	camera:C004	—	172.18.0.1
02 Oct 2026, 22:09:07	admin USER	STREAM_ACCESSED	camera:C005	—	172.18.0.1
02 Oct 2026, 22:09:07	admin USER	STREAM_ACCESSED	camera:C007	—	172.18.0.1
02 Oct 2026, 22:09:07	admin USER	STREAM_ACCESSED	camera:C006	—	172.18.0.1
02 Oct 2026, 22:08:58	admin USER	STREAM_ACCESSED	camera:C003	—	172.18.0.1
02 Oct 2026, 22:08:58	admin USER	STREAM_ACCESSED	camera:C001	—	172.18.0.1
02 Oct 2026, 22:08:58	admin USER	STREAM_ACCESSED	camera:C002	—	172.18.0.1
02 Oct 2026, 22:08:58	admin USER	STREAM_ACCESSED	camera:D001	—	172.18.0.1
02 Oct 2026, 22:08:57	admin USER	LOGIN_SUCCESS	user:1	—	172.18.0.1
02 Oct 2026, 22:08:13	correlation-engine SYSTEM	ALERT_RAISED	alert:924	Detected: GJ01XX2468 Event Id: 91993 Camera Id: C006 Match Type: EXACT Watchlisted: GJ01XX2468 Watchlist Entry Id: 9	
02 Oct 2026, 22:08:13	correlation-engine SYSTEM	ALERT_RAISED	alert:923	Detected: GJ01XX2468 Event Id: 91992 Camera Id: C001 Match Type: EXACT Watchlisted: GJ01XX2468 Watchlist Entry Id: 9	
02 Oct 2026, 22:08:13	correlation-engine SYSTEM	ALERT_RAISED	alert:922	Detected: GJ01XX0001 Event Id: 91991 Camera Id: C006 Match Type: EXACT Watchlisted: GJ01XX0001 Watchlist Entry Id: 1	

Audit log (administrators): every security-relevant action with actor, action, entity, readable details and source IP.

NETRA

CCTV Intelligence Console

Dashboard

Live Wall

Cameras

Alerts751

Detections

Trace

Watchlist

Audit Log

API Clients

Platform Administrator

admin - ADMIN

Sign out

Q Search vehicle / entity (e.g. GJ01XX0001) and trace...

LIVE · 4 msg/min

02 Oct 22:10:31 IST

API clients

Machine identities for AI inference engines, edge gateways and partner VMS (X-API-Key)

ISSUED KEYS

NAME	KEY PREFIX	SCOPES	CREATED	LAST USED	
analytics-simulator	ntr_WnozWb...	events:write, cameras:heartbeat, cameras:write	25 Sep 2026, 20:25:28	38 seconds ago	Revoke

ISSUE NEW KEY

Client name

edge-gateway-ahm-west

Scopes

☒ events:write
 ☒ cameras:heartbeat
 ☐ cameras:write

Issue key

API clients: machine identities with scopes, key prefix, creation and last-use time; a new key is shown exactly once.

Netra CCTV Intelligence Platform

1.0.0 OAS 3.1

/api/openapi.json

Unified CCTV monitoring, video-analytics ingestion, watchlist correlation and real-time alerting.

Authentication

Operators / admins: `POST /api/v1/auth/login` → `Authorization: Bearer <jwt>` (or use `Authorize`).

Machines (AI engines, edge gateways): `X-API-Key: ntr_...` issued via `POST /api/v1/api-clients`.

Real-time: `wss://<host>/api/v1/ws?token=<jwt>` streams `detection.created`, `alert.created`, `alert.updated`, `camera.status` ... messages.

Authorize

auth & users

POST

/api/v1/auth/login

Login

POST

/api/v1/auth/token

Token Form

POST

/api/v1/auth/logout

Logout

GET

/api/v1/auth/me

Me

GET

/api/v1/departments

Departments

GET

/api/v1/users

List Users

POST

/api/v1/users

Create User

GET

/api/v1/api-clients

List Clients

Interactive API documentation (OpenAPI / Swagger UI) at `/api/docs`.

Netra — Technical Handbook · okDriver Full Stack Developer Challenge

14 / 63

3 Running the project

3.1 Requirements

Docker Desktop (or Docker Engine with Compose v2), `make`, `openssl`, about 4 GB of free RAM. Nothing else is installed on the host — every service runs in a container.

3.2 One command

```
git clone https://github.com/AdityaKaushik11/netra.git && cd netra
make up          # generates .env with random secrets, builds the images, starts 7 containers
open https://localhost:8443
```

`make up` runs `scripts/gen-env.sh` first: it copies `.env.example` to `.env` and replaces every secret with a random value from `openssl rand` (database and Redis passwords, JWT secret, Fernet key, gateway secret, the three user passwords, the simulator API key and the simulated devices' credentials), then `chmod 600` s the file. The first start builds four images and takes a few minutes; after 30–60 seconds all cameras report ONLINE and simulated traffic begins.

3.3 Accounts

Three users are seeded — `admin`, `operator`, `viewer` — with the passwords in `.env`. Change one at any time (bcrypt-hashed, audited, and every existing session of that user is signed out):

```
docker compose exec backend python -m app.manage set-password admin
```

3.4 Demo commands

Command	What happens
<code>make scenario</code>	GJ01XX0001 (stolen, CRITICAL) read at C001 → C002 → C005, timestamped like the brief's 10:02 / 10:18 / 10:41 example
<code>make scenario-live</code>	Same route plus C006, one sighting every 8 s so alerts appear one after another on camera
<code>make clone</code>	GJ01XX2468 at two cameras 24 km apart two minutes apart → impossible-travel anomaly
<code>make burst</code>	10 repeat reads of one plate in 3 s → one stored event with repeat count 9
<code>make onboard</code>	API-based bulk onboarding of cameras C007 and C008 (appear live, no refresh)
<code>scripts/camera.sh down c002</code>	Stops a camera's stream → C002 turns OFFLINE within ~20 s; <code>up c002</code> restores it
<code>scripts/camera.sh vendor weak</code>	Vendor NVR reports a weak signal → C005 DEGRADED; <code>offline</code> / <code>online</code> also work
<code>scripts/camera.sh edge pause</code>	Edge agent stops heartbeats → C006 DEGRADED then OFFLINE; <code>resume</code> restores
<code>scripts/camera.sh dashcam pause</code>	Dashcam loses 4G → D001 DEGRADED then OFFLINE
<code>make logs</code> · <code>make ps</code> · <code>make down</code> · <code>make reset</code>	Follow logs · list containers · stop · stop and delete all data
<code>make test</code> · <code>make lint</code>	Backend test suite · all linters and type checks

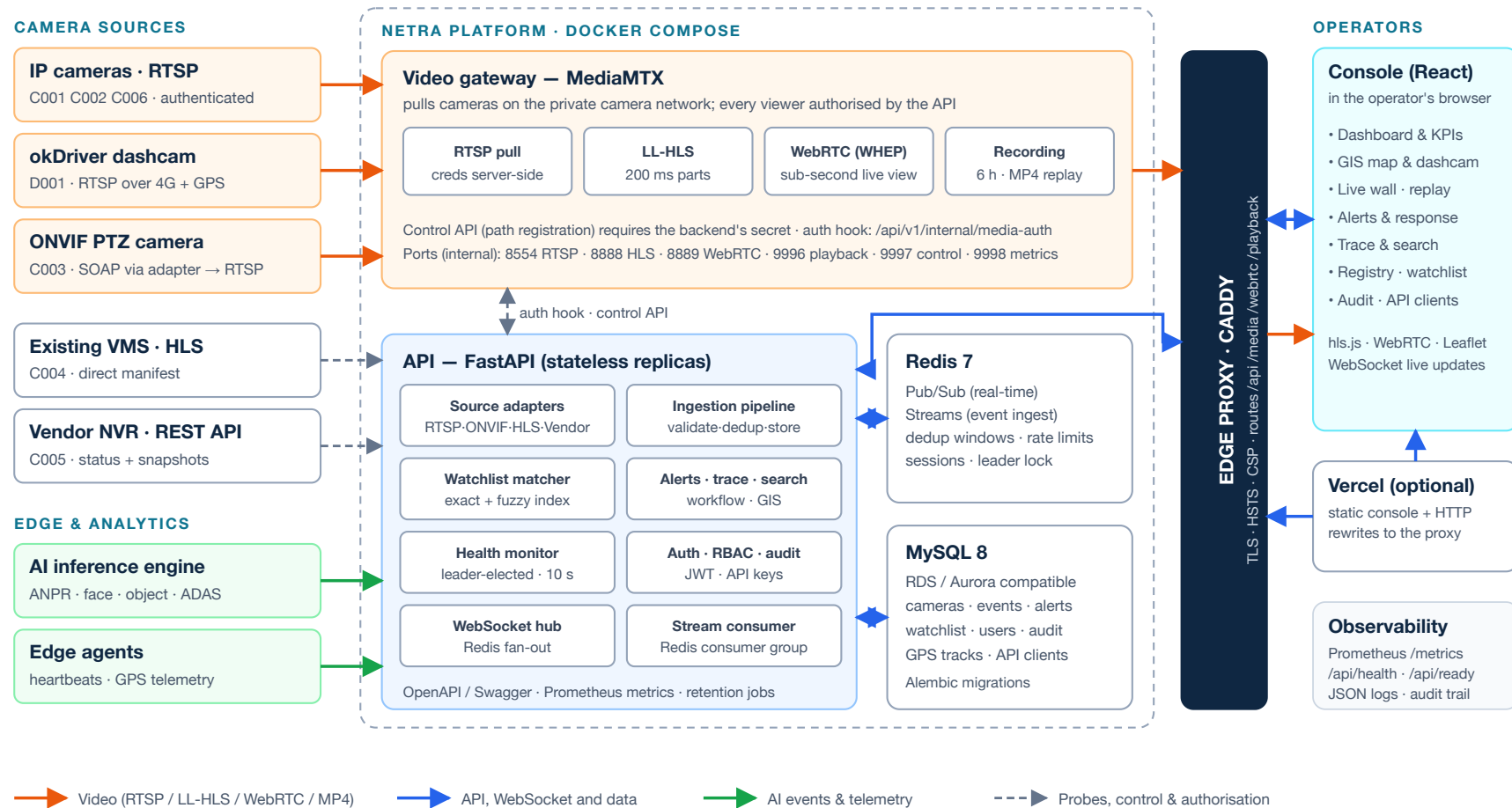
PART II

Architecture

The shape of the system: the components, how they connect, and how the code is organised.

4 System architecture

Sources on the left, the platform in the middle, operators on the right. Video, API/data, AI events and control traffic travel on separate, explicitly authorised paths.



In the prototype the cameras, ONVIF device, vendor NVR, AI engine, edge agent and dashcam are simulators that speak the real protocols; any real RTSP / ONVIF / HLS source is onboarded the same way.

Figure 1 — System architecture

4.1 How to read Figure 1

- **Video** (orange) never passes through the API. The video gateway pulls each camera's RTSP stream on the private camera network and serves it to browsers through the HTTPS proxy. The API only *authorises* each viewer and tells the gateway which cameras exist.
- **Data** (blue) is the REST API and the WebSocket between the console and the API, and the API's connections to MySQL and Redis.
- **Events** (green) are AI inference results and heartbeats arriving from the edge — through REST or through a Redis Stream.
- **Control** (dashed) is the API probing cameras (manifest fetches, ONVIF SOAP, vendor REST) and the gateway asking the API "may this viewer watch this camera?".

4.2 Five architectural decisions

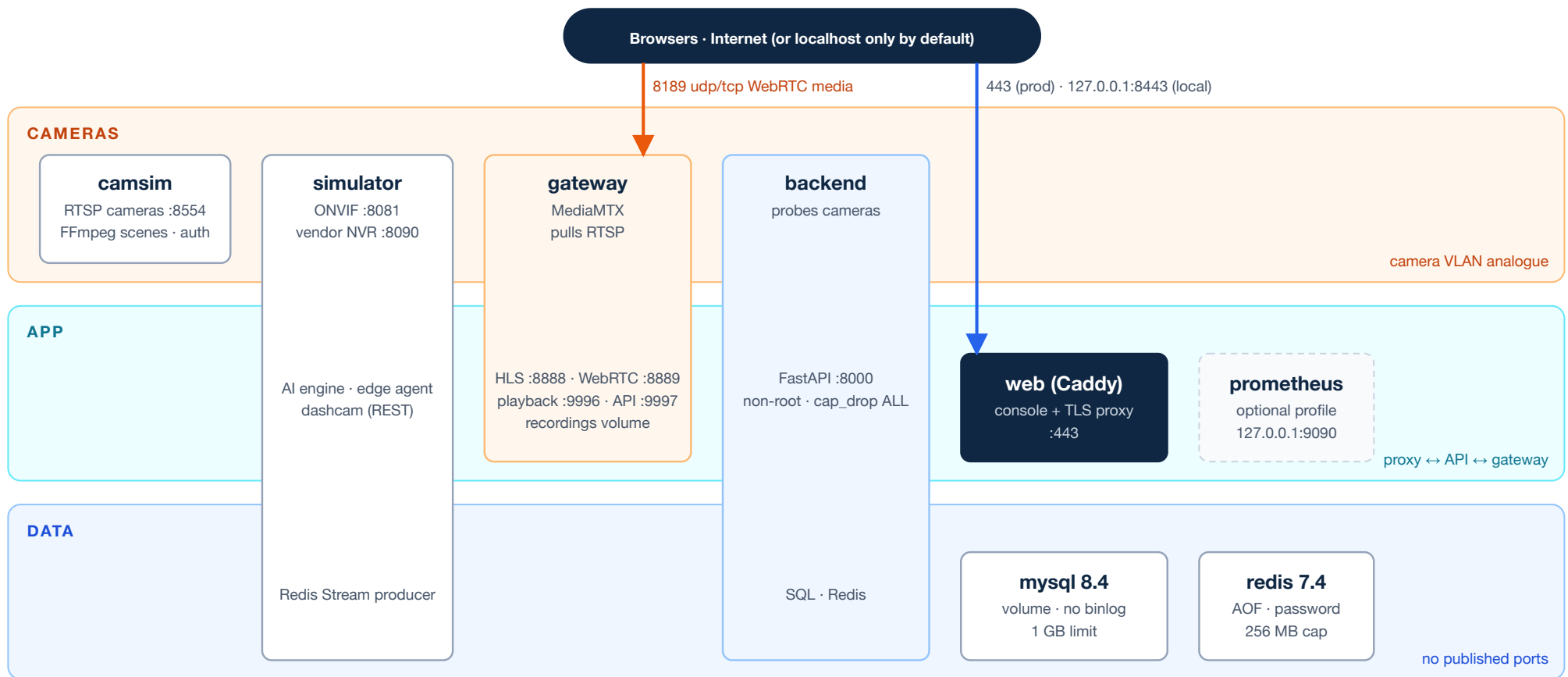
Decision	Why	Consequence
A dedicated video gateway	Browsers cannot play RTSP; cameras must not be exposed; credentials must stay server-side	One component converts RTSP to LL-HLS/WebRTC, records, and serves clips; the API stays lightweight
Adapters per source type	Heterogeneous cameras are the core problem	Registry, health, playback and analytics never branch on vendor
Redis for real-time and dedup	The API must scale horizontally; dedup and rate limits must hold across replicas	Any replica serves any WebSocket; guarantees are global, not per process
MySQL as system of record	okDriver's stack; RDS-compatible; strong indexing for the query paths	Alembic migrations; UTC timestamps; portable schema
Edge proxy in front of everything	One origin for the browser; TLS; security headers; route blocking	Caddy is the only exposed component

5 Repository walkthrough

Where everything lives. The repository has four deployable units (backend, frontend, simulator, infrastructure) plus documentation and scripts.

Path	Purpose
backend/app/main.py	FastAPI application factory: middleware (request id, security headers, CORS, error handler), routers, lifespan that starts the background tasks
backend/app/core/config.py	Every setting, read from environment variables (chapter 7)
backend/app/core/security.py	bcrypt passwords, JWT access and stream tokens, API-key hashing, Fernet encryption
backend/app/core/netguard.py · logging.py · db.py	SSRF guard for every outbound connection to a camera · structured JSON logs · async SQLAlchemy engine and sessions
backend/app/models/	entities.py (9 tables), enums.py (roles, statuses, event types...), types.py (UTC datetime type, BIGINT ids)
backend/app/schemas/	Pydantic request/response models with validation: auth, camera, event, alert/watchlist
backend/app/api/deps.py	Authentication and authorisation dependencies (users, roles, API keys, scopes, rate limits)
backend/app/api/routes/	auth , cameras , events (+ search, trace), alerts , watchlist , system (stats, audit, health, metrics, WebSocket)
backend/app/adapters/	base.py (interface), rtsp.py , onvif.py , http_sources.py (HLS, vendor API)
backend/app/services/	ingestion , watchlist_matcher , dedup , event_bus , stream_consumer , health_monitor , gateway , sessions , ratelimit , audit , serializers , redis_client
backend/app/seed.py · manage.py	Idempotent bootstrap data · operational commands (password reset)
backend/alembic/	Database migrations (initial schema; dashcam telemetry)
backend/tests/	38 automated tests (chapter 27)
frontend/src/api/	Fetch wrapper with auth (client.ts) and all TypeScript types (types.ts)
frontend/src/hooks/	auth (session), realtime (WebSocket and cache updates), location (operator geolocation)
frontend/src/components/	Layout, maps, video player, alert detail, forms, badges, error boundary (chapter 22)
frontend/src/pages/	One file per screen (chapter 23)
simulator/sim/	ONVIF device, vendor NVR, AI analytics engine, edge agent, dashcam, scenarios (chapter 24)
infra/camsim/	Simulated IP cameras: MediaMTX + FFmpeg scenes
infra/gateway · infra/caddy · infra/prometheus	Video gateway configuration · HTTPS edge proxy (routes, headers, CSP, limits) · optional metrics
docker-compose.yml · docker-compose.prod.yml	The stack · production overrides
scripts/	gen-env.sh , camera.sh , deploy-vm.sh , backup.sh , configure-vercel.sh
docs/ · .github/workflows/	Architecture PDF, this handbook, API, security, scalability, deployment, demo script, requirements traceability · CI on every push

6 Runtime: containers, networks, start-up



A container spanning several bands is attached to each of those networks.

All containers: restart unless-stopped · no-new-privileges · memory limit · rotated logs (3 × 10 MB). Backend port 8000 is bound to 127.0.0.1 for local debugging only and removed in production.

Figure 2 — Containers, networks and published ports

6.1 The seven containers (eight with optional Prometheus)

Container	Image	Role
mysql	mysql:8.4	System of record; binary logs off, 256 MB buffer pool, 200 connections, data in a named volume
redis	redis:7.4-alpine	Password-protected; append-only persistence; 256 MB cap evicting only keys with a TTL
backend	built from backend/	API + background tasks; runs migrations and seed at start; non-root user
gateway	bluenviron/mediamtx:1	Video gateway; recordings in a named volume
camsim	built from infra/camsim/	Five simulated RTSP cameras that require credentials
simulator	built from simulator/	ONVIF device, vendor NVR, AI engine, edge agent, dashcam; non-root user
web	built from frontend/	Caddy serving the built React console and proxying the API and video

6.2 Start-up sequence

- 1 MySQL and Redis start; Compose waits until their health checks pass.
- 2 The backend container's entrypoint runs `alembic upgrade head` (creates or migrates the schema), then `python -m app.seed` (idempotent: only creates what is missing), then starts `uvicorn`.
- 3 The FastAPI *lifespan* starts four background tasks in the same process: the real-time subscriber (Redis Pub/Sub → WebSockets), the Redis Stream consumer, the health monitor, and a one-off gateway reconciliation that re-registers every enabled RTSP/ONVIF camera with the gateway.
- 4 The gateway starts with no camera paths; the reconciliation registers them through its control API, authenticated with the gateway secret.
- 5 The simulator waits for a healthy backend, then starts its servers and traffic; the dashcam starts driving.
- 6 Within one or two health cycles (10 s each) every camera has a status.

6.3 Least privilege and limits

Every container: `restart: unless-stopped`, `no-new-privileges`, a memory limit (MySQL 1 GB, backend 768 MB, camsim 768 MB, gateway 512 MB, simulator 384 MB, Redis 320 MB, web 256 MB) and rotated logs (3 × 10 MB). Application containers drop all Linux capabilities; the web container keeps only `NET_BIND_SERVICE` to bind port 443. Locally all ports bind to 127.0.0.1.

PART III

Backend in depth

How each part of the FastAPI service works — configuration, data, security, cameras, health, video, ingestion, matching, alerts, tracing, real-time delivery, statistics, retention and the complete API.

7 Configuration

All configuration comes from environment variables (`pydantic-settings`); nothing secret is in the code. Defaults below are the code defaults; Compose overrides some for the containerised stack.

Setting (env var)	Default	Meaning
DATABASE_URL	sqlite (dev)	Compose sets <code>mysql+aiomysql://netra:...@mysql:3306/netra</code>
REDIS_URL	empty	Empty = in-process fallbacks (tests, single node)
JWT_SECRET	dev value	Signs tokens and keys API-key hashes; refused in production if left at default
ACCESS_TOKEN_MINUTES · STREAM_TOKEN_SECONDS	480 · 300	Session length · camera stream token lifetime
CREDENTIALS_KEY	derived (dev)	Fernet key that encrypts camera credentials
RATE_LIMIT_LOGIN / API / INGEST_PER_MIN	10 · 600 · 6000	Per IP · per user · per API key (production: login 60)
GATEWAY_API_URL · GATEWAY_API_SECRET	gateway:9997 · —	Gateway control API and the backend's credential for it
GATEWAY_PUBLIC_HLS_BASE · ... PLAYBACK_BASE	/media · /playback	URL prefixes handed to browsers
STREAM_ON_DEMAND	false	Pull a camera only while someone watches (bandwidth saving)
HEALTH_INTERVAL_SECONDS	10	Health-monitor cycle
HEARTBEAT_DEGRADED_AFTER_S · ... OFFLINE...	30 · 90	Heartbeat age thresholds
PROBE_TIMEOUT_S · PROBE_SLOW_THRESHOLD_S	3 · 1.5	Probe timeout; slower HLS manifest = DEGRADED (Compose: 6 · 3)
DEDUP_WINDOW_SECONDS	60	Repeat reads of the same identity at the same camera are folded
ALERT_COOLDOWN_SECONDS	300	Open alert for the same entity + camera is updated instead of re-raised
MIN_MATCH_CONFIDENCE · FUZZY_MATCH_ENABLED	0.60 · true	Correlation thresholds
MAX_EVENT_AGE_HOURS · MAX_FUTURE_SKEW_SECONDS	72 · 300	Accepted timestamp window
STREAM_INGEST_ENABLED · STREAM_INGEST_KEY	true · netra:ingest:detections	Redis Stream message channel
EVENT_RETENTION_DAYS · POSITION_RETENTION_HOURS	30 · 48	Data retention
BLOCKED_ENDPOINT_HOSTS	platform services	Hosts that may never be used as camera endpoints
ONVIF_DISCOVERY_HOSTS	empty	ONVIF device URLs probed by discovery
API_DOCS_ENABLED · ENVIRONMENT	true · development	Swagger on/off · production safety checks

Compose-level variables (in `.env`): `HTTPS_PORT` , `BIND_ADDRESS` , `SITE_ADDRESS` , `TLS_MODE` , `HSTS_MAX_AGE` , `EXTERNAL_MEDIA_ORIGINS` , `PUBLIC_IP` , the data-store passwords, the three user passwords, the simulator API key, the simulated devices' credentials, and simulator traffic settings (`INGEST_MODE` , `SIM_EVENTS_PER_MIN` , `PUBLIC_HLS_URL`).

8 Data model, column by column

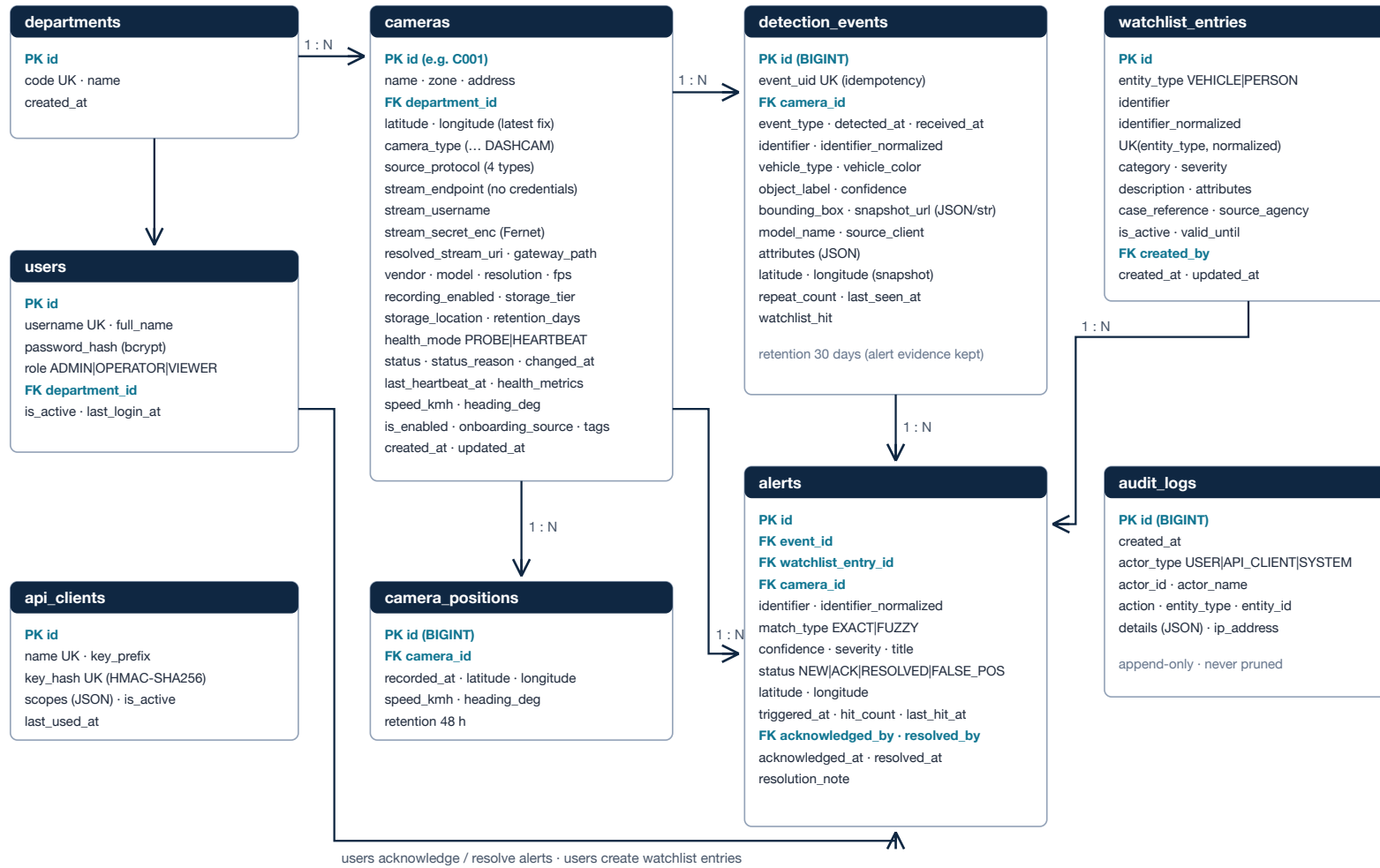


Figure 3 — Entity-relationship diagram (9 tables)

8.1 Conventions

- **Timestamps** are stored as UTC `DATETIME(6)` (microseconds) by a custom `UTCDateTime` type that always returns timezone-aware values, so no code ever compares naive and aware datetimes.
- **Enums** are stored as short strings (not native MySQL ENUMs), so adding a value needs no table rebuild.
- **High-volume ids** (events, audit, GPS points) are BIGINT; camera ids are human-readable strings (`C001` , `D001`).
- **Nothing important is hard-deleted**: cameras are disabled, watchlist entries deactivated; audit logs and alert evidence are kept.

8.2 cameras

Column	Meaning
<code>id</code>	Camera id, e.g. C001; must match <code>^[A-Z0-9][A-Z0-9_-]{1,31}\$</code>
<code>name</code> , <code>zone</code> , <code>address</code> , <code>department_id</code>	Descriptive and organisational metadata
<code>latitude</code> , <code>longitude</code>	Location; for a dashcam, the latest GPS fix
<code>camera_type</code>	FIXED, PTZ, ANPR, DOME, BULLET, THERMAL or DASHCAM
<code>source_protocol</code>	RTSP, ONVIF, HLS or VENDOR_API — selects the adapter
<code>stream_endpoint</code>	Endpoint reference <i>without</i> credentials (rtsp://..., ONVIF device-service URL, manifest URL or vendor API base)
<code>stream_username</code> , <code>stream_secret_enc</code>	Username in clear; password or vendor API key Fernet-encrypted; never returned by the API
<code>resolved_stream_uri</code> , <code>gateway_path</code>	RTSP URI discovered via ONVIF; the gateway path name (<code>cam-c001</code>)
<code>vendor</code> , <code>model</code> , <code>resolution</code> , <code>fps</code>	Device metadata
<code>recording_enabled</code> , <code>storage_tier</code> , <code>storage_location</code> , <code>retention_days</code>	Storage metadata; recording is applied on the gateway
<code>health_mode</code>	PROBE (platform probes the source) or HEARTBEAT (an edge device reports)
<code>status</code> , <code>status_reason</code> , <code>status_changed_at</code>	ONLINE / DEGRADED / OFFLINE / UNKNOWN, a human-readable reason and when it changed
<code>last_heartbeat_at</code> , <code>health_metrics</code>	Last successful probe or heartbeat; metrics such as bitrate, fps, packet loss, temperature
<code>speed_kmh</code> , <code>heading_deg</code> , <code>location_updated_at</code>	Dashcam telemetry
<code>is_enabled</code> , <code>onboarding_source</code> , <code>tags</code>	Disabled cameras keep their history; MANUAL / API / DISCOVERY / SEED

8.3 detection_events

Column	Meaning
<code>event_uid</code>	Unique idempotency key — the producer's <code>event_id</code> , or a generated UUID

camera_id, event_type	Source camera; ANPR, VEHICLE/PERSON/OBJECT_DETECTION, FACE_RECOGNITION, OVERSPEED, HARSH BRAKING, COLLISION_WARNING, DRIVER_DROWSINESS
detected_at, received_at	When the AI saw it (producer clock) and when the platform received it
identifier, identifier_normalized	Plate or person reference as read, and upper-case alphanumerics only (used for matching and search)
vehicle_type, vehicle_color, object_label	Enrichment from the model; ADAS descriptions live in object_label
confidence, bounding_box	0–1 confidence; x, y, w, h of the detection in the frame
snapshot_url, model_name, source_client, attributes	Evidence link, model identity, which API client sent it, extra JSON (e.g. speed, plate-format flag)
latitude, longitude	Snapshot of the location at detection time — the camera's, or the dashcam's GPS fix
repeat_count, last_seen_at	Repeat reads folded into this event by the dedup window
watchlist_hit	True if this event raised or updated an alert

8.4 alerts

Column	Meaning
event_id, watchlist_entry_id, camera_id	What was seen, what it matched, where
identifier, identifier_normalized	As read by the camera (for fuzzy matches this differs from the watchlist identifier)
match_type, confidence, severity, title	EXACT or FUZZY; highest confidence seen; LOW–CRITICAL; human title
status	NEW → ACKNOWLEDGED → RESOLVED or FALSE_POSITIVE
latitude, longitude, triggered_at	Where and when the alert was raised
hit_count, last_hit_at	Further sightings at the same camera within the cool-down
acknowledged_by/at, resolved_by/at, resolution_note	Who handled it, when, and the operator's note

8.5 Other tables

Table	Columns and purpose
watchlist_entries	entity_type (VEHICLE/PERSON), identifier + normalised (unique per type), category (STOLEN_VEHICLE, BLACKLISTED_VEHICLE, WANTED_PERSON, MISSING_PERSON, SUSPICIOUS), severity, description, attributes (make, colour, model, age...), case_reference, source_agency, is_active, valid_until, created_by
users	username (unique), full_name, bcrypt password_hash, role, department, is_active, last_login_at
api_clients	name, key_prefix (first 10 characters, for recognition), key_hash (HMAC-SHA256), scopes, is_active, last_used_at
departments	code and name (AHM-TRF, AHM-CITY, RTO-GJ, GNR-POL in the seed)

camera_positions	Dashcam GPS track: camera_id, recorded_at, latitude, longitude, speed, heading
audit_logs	created_at, actor_type (USER / API_CLIENT / SYSTEM), actor id and name, action, entity type and id, JSON details, IP address

Every index exists for a specific query path; the complete list is in the architecture PDF (section 10). Migrations: `initial schema` and `dashcam telemetry` (adds the speed/heading columns and the `camera_positions` table).

9 Identity, roles and sessions

9.1 Signing in

- 1 `POST /auth/login {username, password}` . The username is trimmed and lower-cased.
- 2 Rate limit: at most 10 attempts per minute per client IP (Redis fixed window).
- 3 If the username has 5 or more recent failures, answer `429` "locked for 15 minutes" and audit `LOGIN_LOCKED` .
- 4 Look up the user and verify the bcrypt hash. If the user does not exist, a dummy hash is checked anyway so the response time does not reveal whether a username exists.
- 5 On failure: increment the failure counter (expires 15 minutes after the last failure), audit `LOGIN_FAILED` , answer `401`.
- 6 On success: clear the counter, record `last_login_at` , audit `LOGIN_SUCCESS` , and return a JWT plus the user profile.

9.2 Tokens

Token	Claims	Use
Access	<code>sub</code> (user id), <code>role</code> , <code>typ=access</code> , <code>iat</code> , <code>exp</code> (+8 h), <code>jti</code> (unique id)	Sent as <code>Authorization: Bearer</code> on every API call and as <code>?token=</code> when opening the WebSocket
Stream	<code>sub</code> , <code>cam</code> (one camera), <code>typ=stream</code> , <code>exp</code> (+5 min)	Appended to video URLs; verified by the gateway through the backend for every request

Tokens are HS256 with `exp` , `sub` and `typ` required. A token of the wrong type is rejected, so a stream token can never be used as a session. On every request the backend also checks revocation: `POST /auth/logout` stores the token's `jti` in Redis until it would have expired; a password reset stores a "revoked before" timestamp for the user, invalidating every older token. The browser keeps the token in `sessionStorage` (cleared when the tab closes).

9.3 Machine identities (API keys)

Administrators issue keys for AI engines, edge agents and dashcams (`POST /api-clients`). A key is `ntr_` + 32 random bytes, shown once; only its HMAC-SHA256 is stored, keyed with the server secret. Each key has explicit scopes:

Scope	Allows
<code>events:write</code>	<code>POST /events</code> , <code>POST /events/batch</code>
<code>cameras:heartbeat</code>	<code>POST /cameras/{id}/heartbeat</code> (including GPS)
<code>cameras:write</code>	<code>POST /cameras</code> , <code>POST /cameras/bulk</code> (API-based onboarding)

Machine requests are rate-limited per key (6,000 per minute); `last_used_at` is updated at most once a minute to avoid a database write per event.

9.4 Who can do what

Action	Viewer	Operator	Admin	API key
See dashboard, cameras, video, detections, alerts, trace, watchlist	✓	✓	✓	—

Acknowledge / resolve alerts (single and bulk)	—	✓	✓	—
Add / edit / deactivate watchlist entries	—	✓	✓	—
Probe a camera now · export detections CSV	—	✓	✓	—
Onboard, edit, enable / disable cameras · ONVIF discovery	—	—	✓	cameras:write (create)
Users, API keys, audit log	—	—	✓	—
Post detection events	—	—	✓	events:write
Send heartbeats / GPS	—	—	✓	cameras:heartbeat

10 Camera registry and source adapters

10.1 Onboarding a camera

- 1 **Validation.** Lengths, coordinate ranges, resolution format, storage tier, and three endpoint rules: it must be an absolute URL; it must not contain inline credentials (`rtsp://user:pass@...` is refused — credentials go in separate fields); its scheme must match the protocol (RTSP → `rtsp/rtsp`, the others → `http/https`). The SSRF guard refuses platform services, loopback and link-local addresses.
- 2 **Encryption.** The password or vendor API key is Fernet-encrypted into `stream_secret_enc`.
- 3 **Adapter prepare.** The camera's adapter makes it usable: RTSP registers a gateway path; ONVIF first resolves the stream URI over SOAP.
- 4 **Audit and broadcast.** `CAMERA_ONBOARDED` is recorded; `camera.created` is pushed to every console; a health probe is scheduled immediately.

Three ways in: the console form (admin), `POST /cameras` or `POST /cameras/bulk` with an API key (up to 500 cameras per call; each one succeeds or fails on its own inside a savepoint), and ONVIF discovery, which pre-fills the form. Editing applies the same rules, refuses `null` for required fields, records a field-level diff (passwords shown as "rotated"), and re-prepares the camera when its endpoint, credentials or recording flag change. Disabling tears down the gateway path and marks the camera OFFLINE "disabled by administrator".

10.2 The adapter interface

```
class CameraAdapter:
    async def prepare(camera)          # on onboarding/edit: register with gateway, resolve URIs
    async def teardown(camera)         # on disable
    async def probe(camera) -> ProbeResult # status + reason + metrics
    def playback(camera, token, exp) -> PlaybackInfo # how a browser renders it (never
credentials)
```

Adapter	prepare	probe	playback
RTSP	Builds the source URL with decrypted credentials and registers gateway path <code>cam-<id></code> (recording on/off)	Reads the gateway path state (section 11.3); re-registers it if the gateway lost it	LL-HLS URL + WebRTC (WHEP) URL + stream token
ONVIF	SOAP <code>GetProfiles</code> → <code>GetStreamUri</code> ; stores the URI; registers the gateway path with the same credentials	<code>GetDeviceInformation</code> (device alive?) then the gateway path state	As RTSP
HLS	—	Fetches the manifest (SSRF-checked, ≤ 3 checked redirects)	The manifest URL itself, no token (external origin)
Vendor API	—	<code>GET {base}/status</code> with <code>X-API-Key</code>	Snapshot URL on the backend + stream token

10.3 ONVIF in detail

ONVIF devices speak SOAP 1.2 over HTTP and authenticate with WS-Security *UsernameToken* digests. For each request the client generates a 16-byte random nonce and a UTC timestamp and sends:

```
PasswordDigest = Base64( SHA-1( nonce + created + password ) )
```

so the password itself never crosses the network. Usernames and profile tokens are XML-escaped; responses are parsed with `defusedxml` (immune to XML entity attacks). Discovery (`POST /onvif/discover` with credentials in the body) probes each configured device URL, returns manufacturer, model, firmware, serial and stream URI, and marks devices already registered. Multicast WS-Discovery is not used because it does not cross container or VLAN boundaries; in production discovery would run on an edge gateway.

10.4 Vendor NVR snapshots

Browsers cannot send an `Authorization` header from an `<img>` tag, so vendor cameras are viewed through `GET /cameras/{id}/snapshot?token=...`. The backend checks that the stream token is valid and scoped to that camera, fetches `{base}/snapshot.jpg` with the vendor key, and relays the bytes only if the response really is an image under 5 MB. The console refreshes the image once per second — a near-live, low-bandwidth ($\approx$ 50–80 kbps) mode suited to cameras on weak links.

11 Camera health monitoring

11.1 The loop

The health monitor runs inside every API replica, but only one does the work: each cycle it tries to take (or renew) the Redis key `netra:leader:health-monitor` with `SET NX EX 30`. The leader then evaluates all enabled cameras concurrently (at most 32 probes at a time), saves results, and publishes `camera.status` for every camera whose status changed. A cycle runs every 10 seconds; roughly once an hour it also runs data retention (chapter 19).

11.2 Two health modes

Mode	How status is decided
PROBE	The platform actively checks the source through its adapter (11.3). A successful probe updates <code>last_heartbeat_at</code> .
HEARTBEAT	An edge device reports via <code>POST /cameras/{id}/heartbeat</code> (C006's edge agent every 10 s; the dashcam every 3 s with GPS). No heartbeat yet → OFFLINE; age > 90 s → OFFLINE "heartbeat lost"; age > 30 s → DEGRADED "heartbeat late"; packet loss > 5 % or fps < 5 → DEGRADED; otherwise ONLINE — but for a gateway camera the video is checked too, and if the gateway cannot pull it the camera is DEGRADED "heartbeat OK, video unavailable". A heartbeat only revives an OFFLINE/UNKNOWN camera immediately; every other transition is decided by the monitor, so the two paths never fight.

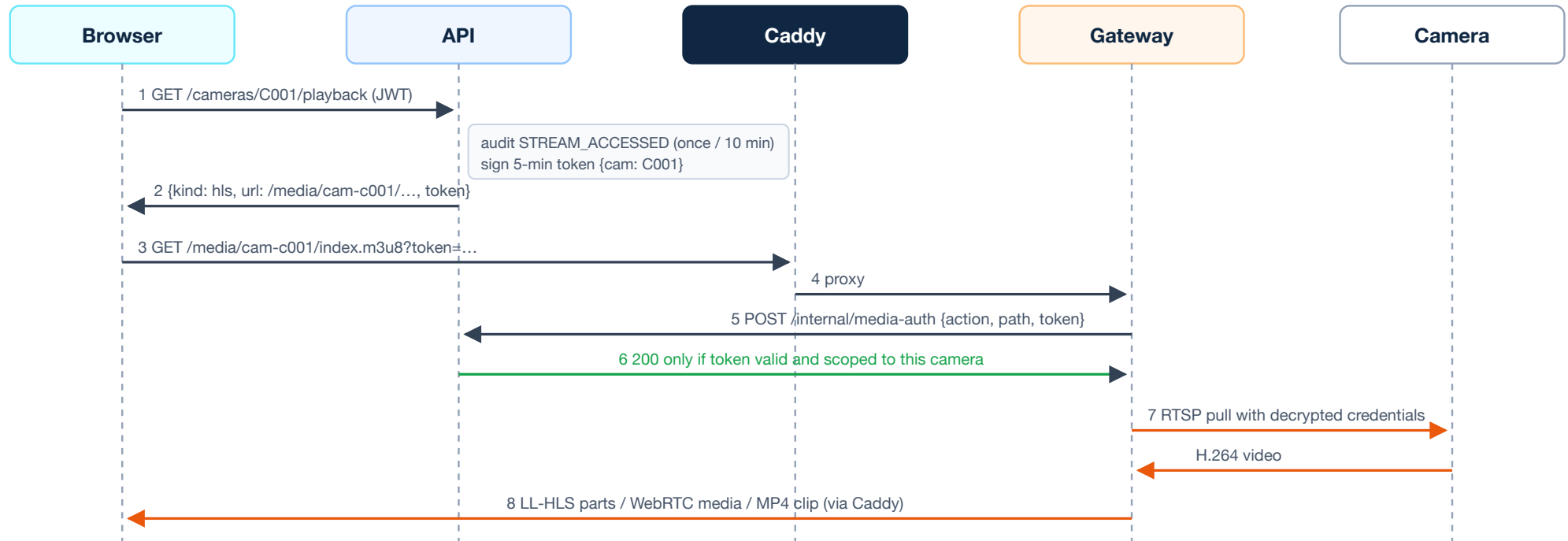
11.3 What each probe checks

Source	ONLINE	DEGRADED	OFFLINE
RTSP / ONVIF	Gateway path ready and inbound bitrate ≥ 40 kbps (bytes received between two probes)	Ready but bitrate below 40 kbps	Path missing (re-registered automatically) or gateway cannot pull the source; ONVIF device unreachable or rejects credentials. Gateway itself unreachable → UNKNOWN
HLS	Manifest 200 with <code>#EXTM3U</code>	Manifest slower than the threshold	Unreachable, non-200 or not a manifest
Vendor API	<code>online: true</code>	Weak signal or fps < 5	Unreachable, credentials rejected, or <code>online: false</code>

11.4 Hysteresis — no flapping

A single failed probe of a healthy camera does not turn it red. In PROBE mode the first OFFLINE result of an ONLINE/DEGRADED camera is recorded as DEGRADED "probe failed, retrying"; only the second consecutive failure makes it OFFLINE. Recovery is immediate. Every change is audited (`CAMERA_STATUS_CHANGED` with from, to and reason), timestamped, exported as the Prometheus gauge `netra_cameras{status}`, and pushed to every console, which shows a toast ("C002 went OFFLINE") and recolours the marker.

12 Video: live, WebRTC, recording



The browser only ever sees a gateway path and a short-lived, camera-scoped token. Camera credentials are stored Fernet-encrypted, never returned by the API, and inline user:pass@ URLs are rejected. The same check guards WebRTC (WHEP) and recorded playback.

Figure 4 — How a browser is allowed to watch a camera

12.1 The gateway

MediaMTX pulls each registered RTSP source over TCP and serves it in three ways: **LL-HLS** (Low-Latency HLS: 1-second segments split into 200 ms parts, seven segments kept, about 1–2 s glass-to-glass), **WebRTC** through the WHEP protocol (sub-second), and **recording** to fragmented MP4 (60-second segments, deleted after 6 hours) served back as MP4 clips by its playback server. Its control API (port 9997) registers paths; its metrics are on 9998.

12.2 Authorising every viewer

The gateway is configured with `authMethod: http` : for every read, playback and API request it calls `POST /api/v1/internal/media-auth` on the backend with the action, path, query string and any credentials. The backend allows:

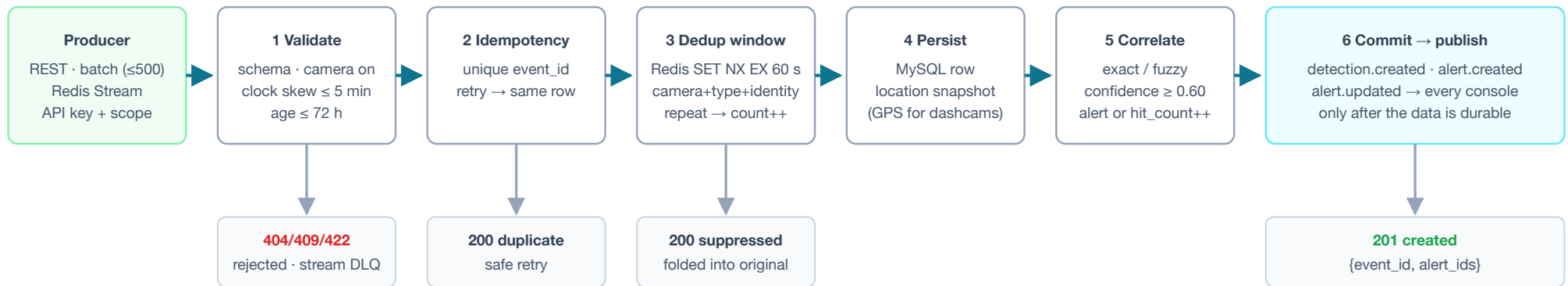
- **read / playback** only if a valid stream token (from the query string or bearer header) names the camera whose path is being requested (`cam-` + lower-cased camera id);
- **api** only if the basic-auth credentials equal `netra-backend` and `GATEWAY_API_SECRET` (constant-time comparison);
- nothing else. The proxy never routes `/api/v1/internal/*` from outside.

12.3 In the browser

Mode	How the console plays it
LL-HLS	<code>hls.js</code> in low-latency mode (live sync 2 parts, 10 s back-buffer). Its request hook re-opens each same-origin request with <code>?token=...</code> appended, so the token reaches the playlist, parts and segments but is never sent to a third-party origin. On a fatal error it retries every 5 s. Tokens are renewed every ~3.3 minutes without restarting playback.
WebRTC	Creates an <code>RTCPeerConnection</code> with a receive-only video transceiver, gathers ICE candidates for up to 1.5 s, POSTs the SDP offer to <code>/webrtc/cam-.../whep?token=...</code> , applies the answer. If it is not connected within 8 s it falls back to LL-HLS automatically. The session is DELETED when the player closes.
Snapshots	An <code></code> refreshed every second from the snapshot proxy (10.4).
Recorded clip	<code>GET /cameras/{id}/clip?start&duration</code> returns a playback URL such as <code>/playback/get?path=cam-c001&start=...&duration=40&format=mp4</code> plus a token; the browser plays the MP4 in a normal <code><video></code> element. Each request is audited as <code>RECORDING_ACCESSED</code> .

Opening a live view is audited as `STREAM_ACCESSED` once per user per camera per 10 minutes — enough to answer "who watched this camera and when" without recording every token renewal.

13 The ingestion pipeline



Measured: 10 repeat reads of one passing vehicle → 1 stored event with repeat_count = 9; a retried POST returns the original event; a watchlisted plate appears as an alert on every open console in under a second.

Figure 5 — Ingestion stages and outcomes

13.1 Stage by stage

The same function, `ingestion.ingest()`, serves `POST /events`, every item of `POST /events/batch` and every message from the Redis Stream.

- 1 Schema validation** (before the pipeline): event type known; timestamp with a timezone; confidence 0–1; ANPR needs a plate of 4–12 alphanumeric after normalisation; face recognition needs a `person_ref`; GPS latitude and longitude together or not at all; attributes under 4 KB; snapshot URL `http(s)` or relative. Errors return 422 with the failing field.
- 2 Semantic validation**: the camera must exist (404) and be enabled (409); the timestamp may be at most 5 minutes in the future (clock skew) and at most 72 hours old (422).
- 3 Idempotency**: if the producer sent an `event_id` that is already stored, answer 200 *duplicate* with the original event id. A producer can therefore retry safely after a timeout.
- 4 Temporal de-duplication**: for events with an identity, try `SET netra:dedup:{camera}:{type}:{identity}<event_uid> NX EX 60` in Redis. If the key already exists, another read of the same vehicle at the same camera arrived within 60 s: the original event's `repeat_count` is incremented, `last_seen_at` moves forward, the higher confidence is kept, and the answer is 200 *suppressed*. Because the key lives in Redis this works across replicas.
- 5 Persist**: the event is inserted with the camera's location — or the GPS position the dashcam sent with it — so history stays correct even if a camera is moved later. ANPR events get a `plate_format_valid` flag (Indian `GJ01AB1234` or BH-series pattern); a dashcam's speed is stored in attributes.
- 6 Correlate**: if the event has an identity, maps to an entity type (vehicle for ANPR / vehicle detection, person for face / person detection) and has confidence ≥ 0.60 , the watchlist matcher runs (chapter 14) and alerts are raised or updated (chapter 15).
- 7 Commit, then publish**: only after the transaction commits are `detection.created`, `alert.created` and `alert.updated` published — consoles never see data that does not exist. The answer is 201 *created* with the event id and alert ids.

13.2 Batch and stream inputs

Batch (≤ 500 events) exists for edge gateways that buffer during a network outage: each event is processed independently and the response lists each result plus totals (created, duplicates, suppressed, rejected). Because of idempotency, replaying a buffer after reconnecting is safe. **Stream**: producers `XADD netra:ingest:detections * payload <json> client <name>`; chapter 17 explains how the consumer group processes it.

13.3 Measured behaviour

Input	Result
10 reads of one passing vehicle within 3 s (make burst)	1 event, repeat_count = 9 (1 × 201, 9 × 200 suppressed)
The same <code>event_id</code> posted twice	One row; second answer 200 duplicate with the same id
Unknown camera / naive timestamp / confidence 1.7 / timestamp 1 h ahead	404 / 422 / 422 / 422
Watchlisted plate at confidence 0.3	Stored, no alert

Metrics: `netra_events_total{event_type,result}`, `netra_alerts_total{severity,match_type}` and the latency histogram `netra_ingest_seconds`.

14 Watchlist matching algorithms

14.1 Normalisation

Both watchlist identifiers and detections are normalised the same way: upper-case, then every character that is not A–Z or 0–9 removed. `gj 01 xx-0001`, `GJ01 XX 0001` and `GJ01XX0001` are the same plate.

14.2 The in-memory index

Matching runs on every detection, so it cannot query the database each time. The matcher keeps all *active* entries in memory:

- an **exact index**: a dictionary from (entity type, normalised identifier) to the entry — one lookup per detection;
- a **deletion index** for vehicles (14.3).

The index is rebuilt when the watchlist changes or after 5 minutes. Every add, edit or deactivation increments the Redis counter `netra:watchlist:version`; each replica compares it with the version it loaded before matching, so no replica uses a stale list for more than one event. Entries past `valid_until` never match.

14.3 Fuzzy matching with a deletion index (SymSpell)

ANPR makes predictable OCR mistakes: `0↔O`, `8↔B`, `5↔S`, `1↔I`, a dropped or extra character. Netra treats a read as a *possible match* if it is within **edit distance 1** (one substitution, insertion or deletion) of a watchlisted plate of at least 6 characters. Comparing each read with every entry would cost $O(\text{watchlist size})$. Instead, for every entry the index stores the plate itself and every string obtained by deleting one character. For a read, the same deletions are generated; any entry that shares a key is a candidate, and candidates are confirmed with an exact edit-distance-1 check. The cost is proportional to the plate length (about 10 lookups), not to the size of the watchlist.

```
Watchlist entry  GJ05XX1234  deletions include  GJ05XX134  (deleted "2")
Camera read      GJ05XX1Z34  deletions include  GJ05XX134  (deleted "Z")
shared key "GJ05XX134" → candidate → one substitution (2→Z) → FUZZY match
```

Fuzzy matches are labelled *possible match*, lowered one severity level (CRITICAL → HIGH, HIGH → MEDIUM...), and the alert view shows both the read and the watchlisted plate so the operator can verify visually. Exact matches always win: if an exact match exists, fuzzy matching is not attempted. Persons (face-recognition gallery references) are matched exactly only.

15 Alerts

15.1 Raise or update

For every match the pipeline first looks for an **open** alert (NEW or ACKNOWLEDGED) for the same watchlist entry at the same camera whose last hit is within the cool-down (300 s). If one exists it is updated — `hit_count + 1`, `last_hit_at`, best confidence — and `alert.updated` is published. Otherwise a new alert is created with the entry's severity (lowered for fuzzy matches), the event's location, a title such as "Stolen vehicle GJ01XX0001 at Paldi Cross Roads Junction", `ALERT_RAISED` is audited by the system actor `correlation-engine`, and `alert.created` is published. A vehicle that keeps sitting in front of one camera therefore produces one alert with a growing hit count; the same vehicle at a different camera produces a new alert, because that is new information.

15.2 Workflow

Action	Rules
Acknowledge	Only from NEW (409 otherwise); optional note; records who and when; audited <code>ALERT_ACKNOWLEDGED</code>
Resolve	From NEW or ACKNOWLEDGED; note of at least 3 characters required; <code>false_positive</code> flag chooses <code>FALSE_POSITIVE</code> ; an unacknowledged alert is acknowledged at the same moment; audited <code>ALERT_RESOLVED</code>
Bulk	<code>POST /alerts/bulk</code> with a list of ids or "all open alerts older than N minutes"; up to 5,000; each alert audited individually

15.3 What the operator is told

The toast (CRITICAL stays 20 s, others 10 s) and a synthesised alarm tone (three beeps for CRITICAL, two for HIGH, one otherwise; can be muted) link straight to the alert and to the trace. The alert view assembles the watchlist record, the event, the camera, the trace and the recording, and adds two automatic checks: a **colour mismatch** warning when the vehicle colour seen differs from the watchlist record (a possible false plate), and a **non-standard plate format** warning from the ingestion flag.

15.4 The response map

The browser's Geolocation API supplies the operator's position (the site allows geolocation for itself only); if it is denied or unavailable the map uses the Ahmedabad control room (Shahibaug) and says so. The wanted vehicle is placed at its *latest* sighting — which may be newer than the alert. Distance is the great-circle distance; the drive time assumes 30 km/h city traffic. "Directions" opens Google Maps with both points; up to six sightings from the previous two hours are drawn as numbered dots.

16 Search and movement tracing

16.1 Search

`GET /search?q=` returns three groups: detected entities whose normalised identifier contains the text (with number of detections, cameras and last seen), watchlist entries containing it, and cameras whose id or name contains it. Input that normalises to nothing (for example `--`) does not match the whole watchlist. The top bar's search box jumps straight to a trace.

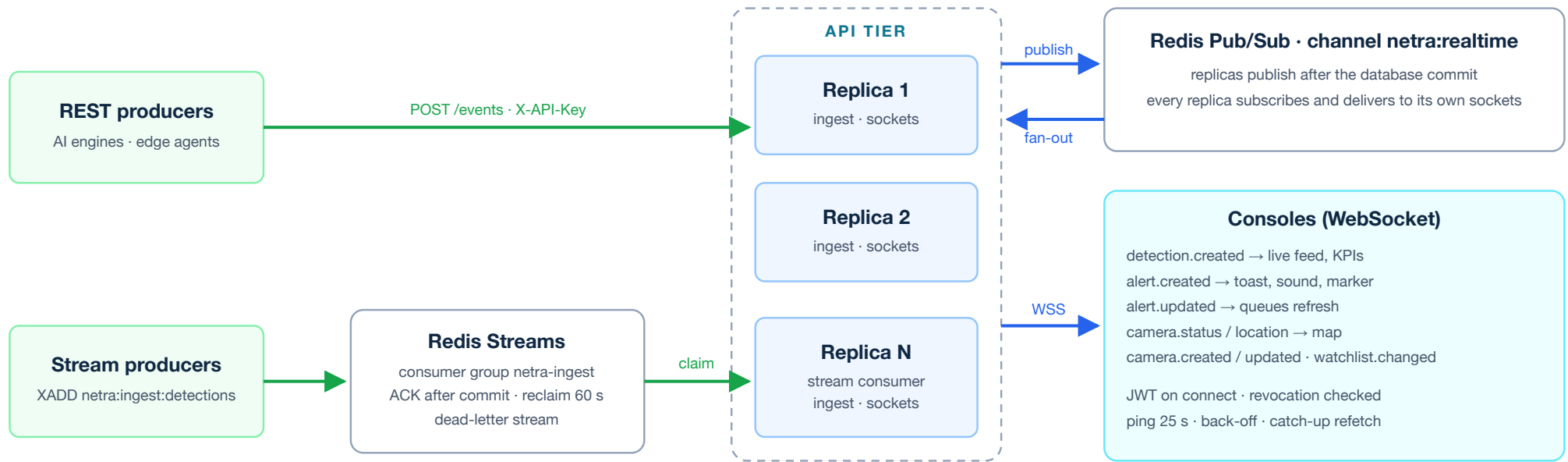
16.2 Trace

`GET /trace/{identifier}` loads the 2,000 most recent sightings of the normalised identifier (optionally within a time range), sorts them oldest first and computes, for each stop: minutes since the previous sighting, the great-circle (haversine) distance, and the implied speed.

$$d = 2 \cdot 6371 \text{ km} \cdot \text{asin}(\sqrt{\sin^2(\Delta\phi/2) + \cos \phi_1 \cdot \cos \phi_2 \cdot \sin^2(\Delta\lambda/2)}) \quad \text{speed} = d / \Delta t$$

If two consecutive sightings are more than 1 km apart and either share a timestamp or imply a speed above **160 km/h**, the trace adds the anomaly "Impossible travel A→B: x km in y min — possible cloned/duplicate plate". The response also includes the first and last sighting, totals and, if the identity is on the watchlist, its record and number of alerts. The console draws the stops as a numbered dashed route on the map.

17 Real-time delivery



Any replica can publish; every replica delivers to its own sockets — no sticky sessions, so the API tier scales behind any load balancer. Without Redis (tests) an in-process bus is used.

Figure 6 — Real-time fan-out and the message-channel ingest path

17.1 WebSocket protocol

The console opens `wss://<host>/api/v1/ws?token=<access token>`. The server verifies the token (and that it is not revoked) and that the user is active — otherwise it closes with code 4401 — then sends `{"type":"hello"}`. The client sends `ping` every 25 s and receives `{"type":"pong"}`; a socket silent for 90 s is closed. Every message has the shape `{"type", "ts", "data"}`:

type	data	Console reaction
detection.created	event	Added to the live detections feed; KPIs, history and charts refreshed
alert.created	alert	Toast, sound, alert queues, camera markers, trace and watchlist refreshed
alert.updated	alert / {bulk}	Queues and markers refreshed
camera.status	camera	Toast for OFFLINE / DEGRADED / back ONLINE; map and health refreshed
camera.location	id, lat, lng, speed, heading, at	Dashcam marker moved and trail extended in place (no re-fetch)
camera.created / updated / bulk_created	camera / {ids}	Registry and map refreshed
watchlist.changed	{id, action}	Watchlist refreshed

On the server, `event_bus.publish()` sends every message to the Redis channel `netra:realtime`; every replica subscribes and delivers to its own sockets, dropping any socket that does not accept a message within 2 s. If Redis publishing fails the replica still delivers locally. On the client, cache invalidations are batched (one flush per 1.2 s), and after a reconnect (exponential back-off up to 15 s) everything is re-fetched once to catch up.

17.2 The Redis Stream consumer

Each replica joins the consumer group `netra-ingest` on the stream `netra:ingest:detections` (consumer name = host name) and reads up to 100 messages at a time, blocking for at most 1 s. For each message it parses and validates the payload and runs the same ingestion pipeline; it then acknowledges the message (`XACK`). Invalid messages are copied to the dead-letter stream `netra:ingest:dlq` (capped at 10,000) with the error, then acknowledged. Messages that fail for any other reason (for example the database is briefly down) are **not** acknowledged; every 30 s the consumer claims messages that have been pending for more than 60 s (`XAUTOCLAIM`) — including those of a replica that crashed — and retries them. This gives at-least-once delivery; idempotency makes it effectively exactly-once.

18 Statistics, metrics and audit

18.1 Dashboard statistics

`GET /stats/overview` computes, in one call: cameras by status (enabled only) plus disabled and total; open alerts by severity, NEW alerts and alerts in 24 h; events in the last 5 minutes, hour and 24 hours, events per minute, and repeat reads suppressed in 24 h; active watchlist size and the matcher's cached size; WebSocket clients on this replica; and the platform's own service health (API, database, Redis, ingest backlog from the consumer group's pending + lag, gateway reachability, ready streams and viewers — cached for 5 s). `GET /stats/timeseries?hours=24` returns hourly detections, watchlist hits and alerts.

18.2 Health and metrics endpoints

Endpoint	Purpose
<code>/api/health</code>	Liveness: the process answers (used by Docker's health check)
<code>/api/ready</code>	Readiness: database and Redis reachable (503 otherwise)
<code>/api/metrics</code>	Prometheus metrics; internal only (blocked at the proxy)

18.3 The audit log

Twenty-one actions are recorded, each with the actor (user, API client or system), the entity, readable details and the client IP:

Area	Actions
Sessions	LOGIN_SUCCESS, LOGIN_FAILED (with consecutive failures), LOGIN_LOCKED, LOGOUT, PASSWORD_RESET
Users & keys	USER_CREATED, API_CLIENT_CREATED, API_CLIENT_REVOKED
Cameras	CAMERA_ONBOARDED, CAMERA_UPDATED (field-level diff, secrets masked), CAMERA_ENABLED, CAMERA_DISABLED, CAMERA_STATUS_CHANGED
Video	STREAM_ACCESSED, RECORDING_ACCESSED
Alerts	ALERT_RAISED, ALERT_ACKNOWLEDGED, ALERT_RESOLVED
Watchlist	WATCHLIST_ADDED, WATCHLIST_UPDATED, WATCHLIST_DEACTIVATED

No endpoint updates or deletes audit records, and retention never prunes them.

19 Data retention

A platform that runs for months must not fill its disk. Retention is automatic.

Data	Kept for	Mechanism
Detection events	30 days	Hourly job deletes older events in batches of 5,000 — except events referenced by an alert, which are evidence
Dashcam GPS points	48 hours	Same hourly job
Recordings	6 hours	Gateway <code>recordDeleteAfter</code>
Alerts, watchlist, users, audit	Indefinitely	Never pruned
Redis	By TTL	Dedup keys 60 s, rate windows 1 min, lockouts 15 min, revocations until token expiry; streams length-capped
Container logs	30 MB / container	Docker rotation 3 × 10 MB
MySQL binary logs	None	Disabled (single server; backups instead)
Backups (server)	7 days	Nightly <code>mysqldump --single-transaction</code> , gzipped

20 API reference

Base path `/api/v1` ; JSON; OpenAPI at `/api/docs` . Lists return `{items, total, limit, offset}` ; errors `{"detail": ...}` . "any" = any signed-in user.

Method	Path	Access	Purpose
POST	<code>/auth/login</code> · <code>/auth/token</code>	public	Sign in (JSON / OAuth2 form for Swagger)
POST	<code>/auth/logout</code>	any	Revoke the current token
GET	<code>/auth/me</code> · <code>/departments</code>	any	Current user · departments
GET · POST	<code>/users</code>	admin	List · create users
GET · POST	<code>/api-clients</code>	admin	List · issue API keys
POST	<code>/api-clients/{id}/revoke</code>	admin	Revoke a key
GET	<code>/cameras?q&status&department_id&zone&protocol&camera_type&enabled</code>	any	Search / filter the registry
POST	<code>/cameras</code> · <code>/cameras/bulk</code>	admin · cameras:write	Onboard one · up to 500
GET · PATCH	<code>/cameras/{id}</code>	any · admin	Read · edit / enable / disable
GET	<code>/cameras/zones</code> · <code>/cameras/{id}/audit</code>	any	Zones · camera audit history
POST	<code>/cameras/{id}/heartbeat</code>	cameras:heartbeat	Edge heartbeat with metrics and optional GPS
GET	<code>/cameras/{id}/track?minutes</code>	any	Dashcam GPS track
POST	<code>/cameras/{id}/probe</code>	operator	Probe now
GET	<code>/cameras/{id}/playback</code>	any	Live-view descriptor + stream token
GET	<code>/cameras/{id}/snapshot?token</code>	stream token	Vendor snapshot proxy
GET	<code>/cameras/{id}/recordings</code> · <code>/clip?start&duration</code>	any	Recorded segments · MP4 clip descriptor
POST	<code>/onvif/discover</code>	admin	ONVIF discovery (credentials in body)
POST	<code>/events</code> · <code>/events/batch</code>	events:write · admin	Ingest one · up to 500
GET	<code>/events?camera_id&event_type&identifier&since&until&watchlist_hit&min_confidence</code>	any	Detection history
GET	<code>/events/{id}</code> · <code>/events/export.csv</code>	any · operator	One event · CSV ($\leq 50,000$ rows)
GET	<code>/search?q</code> · <code>/trace/{identifier}</code>	any	Search · movement history
GET · POST	<code>/watchlist</code>	any · operator	List · add

PATCH · DELETE	/watchlist/{id}	operator	Edit · deactivate
GET	/alerts?status&severity&camera_id&identifier&since · /alerts/{id}	any	Alert queue · detail
POST	/alerts/{id}/acknowledge · /resolve · /alerts/bulk	operator	Alert workflow
GET	/stats/overview · /stats/timeseries	any	Dashboard statistics
GET	/audit	admin	Audit log
WS	/ws?token	any	Real-time channel
GET	/api/health · /api/ready	public	Liveness · readiness

Worked example: one detection end to end

```
POST /api/v1/events
X-API-Key: ntr_...
{ "event_id": "edge-42-000187", "camera_id": "C001", "event_type": "ANPR",
  "timestamp": "2026-10-02T10:02:11+05:30", "vehicle_number": "GJ 01 XX 0001",
  "confidence": 0.94, "vehicle_type": "car", "vehicle_color": "white",
  "bounding_box": {"x": 412, "y": 388, "w": 176, "h": 44} }

201 Created
{ "status": "created", "event_id": 91988, "event_uid": "edge-42-000187", "alert_ids": [919] }

WebSocket → every console:
{ "type": "alert.created", "ts": "...", "data": { "id": 919, "identifier": "GJ01XX0001",
  "severity": "CRITICAL", "match_type": "EXACT", "camera_id": "C001", "title": "Stolen vehicle
GJ01XX0001 at Paldi Cross Roads Junction", ... } }
```

PART IV

Frontend in depth

How the React console is built: data flow, real-time updates, the video and map components, and every page.

21 Frontend architecture

Concern	Choice and how it is used
Framework	React 19 with TypeScript in strict mode, built by Vite; routing with React Router 7
Server state	TanStack Query: every screen reads through cached queries (keys such as <code>["cameras", filters]</code> , <code>["alerts", ...]</code> , <code>["overview"]</code>); data is considered fresh for 10 s; mutations invalidate the affected keys
API client	<code>api()</code> wraps <code>fetch</code> : prefixes <code>/api/v1</code> , adds the bearer token, serialises JSON, turns FastAPI validation errors into readable messages, and calls the sign-out handler on a 401
Session	<code>AuthProvider</code> : token in <code>sessionStorage</code> , current user, <code>login</code> , <code>logout</code> (revokes server-side), <code>can(...roles)</code> used to show or hide actions
Real-time	<code>RealtimeProvider</code> owns one WebSocket per tab (chapter 17), keeps the last 80 detections, plays alert tones, shows toasts, updates the dashcam position directly in the query cache, and batches invalidations
Styling	Tailwind CSS 4 with a design-token theme (ink greys, cyan accent, status and severity colours); component classes (panel, input, btn, plate, data table) in <code>index.css</code>
Libraries	Leaflet + react-leaflet (maps), hls.js (video), Recharts (trend chart), sonner (toasts), lucide (icons), date-fns; fonts bundled locally (Inter, JetBrains Mono)
Resilience	An error boundary around the whole app shows "Something went wrong" with Try again / Reload instead of a blank page; routes are protected (unauthenticated → login; non-admin → dashboard for admin pages)

Data flow in one sentence

A screen reads data through a query; the server pushes a WebSocket message when something changes; the real-time provider invalidates (or directly patches) the matching queries; the screen re-renders with fresh data — no polling loops and no page refreshes.

22 Key components

Component	What it does
Layout	Sidebar navigation (admin-only items hidden for others; NEW-alert badge), top bar with entity search, live-connection indicator ("LIVE · n msg/min" or "RECONNECTING"), sound mute, IST clock and the routed page
VideoPlayer	Asks <code>/playback</code> for a descriptor and renders HlsView, WhepView (with automatic HLS fallback) or SnapshotView; overlays camera id, name, status, open-alert count, transport and dashcam speed; refreshes the stream token every 200 s
ClipPlayer	Asks <code>/clip</code> for a recorded window and plays the MP4 with controls
CameraMap	Leaflet map that fits all cameras (or a trace route): status dots, pulsing ring for open alerts, heading arrow for dashcams, purple GPS trails, numbered route stops with times
OsmTiles	OpenStreetMap base layer darkened with a CSS filter; tiles alone send the site origin as referrer (OSM's usage policy)
ResponseMap	Operator position (with accuracy circle) and the wanted vehicle's latest sighting, a dashed line between them, earlier sightings
AlertDetail	The full incident view (section 15.3–15.4), plus acknowledge / resolve with note
AlertToast	Severity-coloured toast with plate, category, camera, time, confidence, case reference, "Open alert" and "Trace movement"
CameraForm	Create / edit camera with protocol-specific hints; password field never pre-filled ("leave blank to keep")
DetailsView	Turns audit and metric JSON into readable chips ("DEGRADED → ONLINE", "Packet Loss %: 0.7")
Badges	Status, severity, alert-status and protocol badges; number-plate styling; confidence bar
useMyLocation	Browser geolocation with permission handling and the control-room fallback

23 Pages

Page	Route	Behaviour
Login	/login	Section 2.1; redirects to the dashboard once signed in
Dashboard	/	Section 2.2; camera filter applies to map, grid and health table; dashcam trails loaded per dashcam
Live wall	/wall	All enabled cameras; 2/3/4 columns; WebRTC toggle remounts players in low-latency mode
Cameras	/cameras	Paged registry with five filters and a map; <code>?focus=ID</code> opens the camera detail modal (live view, WebRTC toggle, recorded replay, dashcam telemetry and track, health, actions, audit); "Add camera" and "ONVIF discovery" for admins
Alerts	/alerts	Five tabs, severity filter, paging, multi-select bulk actions; <code>?focus=ID</code> opens AlertDetail; new alerts flash red
Detections	/events	Filters including a date-time range; CSV export (operators); clicking a row opens its trace
Trace	/trace/:id	Exact trace with map; "Partial-match search" lists entities, watchlist entries and cameras containing the text
Watchlist	/watchlist	Search, category and active filters; add entry (vehicle or person); deactivate; jump to trace
Audit log	/audit	Admins only; filter by entity type and action; refreshes every 15 s
API clients	/api-clients	Admins only; issue a key with scopes (shown once with a copy button); revoke

PART V

Simulators and infrastructure

The simulated city that exercises the platform with real protocols, and the infrastructure configuration that runs it.

24 The simulated world

The brief allows simulated sources. They are built to speak the *real* protocols, so the platform code is the code that would talk to real cameras and models.

24.1 Cameras and devices

Simulator	What it is
camsim (RTSP cameras)	A MediaMTX instance that requires credentials to read, fed by FFmpeg. Each stream (<code>c001</code> , <code>c002</code> , <code>c003</code> , <code>c006</code> , <code>d001</code>) is a generated CCTV-style scene — road, lane markings, moving vehicles, sensor noise — with an on-screen display of the camera id, location and live IST clock; the dashcam stream is a forward-facing road view. H.264 baseline, 960×540, 700 kbps. Dropping <code><stream>.mp4</code> into <code>samples/</code> replaces a scene with real footage.
ONVIF device	SOAP 1.2 endpoint that verifies WS-Security digests exactly like a real camera and answers <code>GetDeviceInformation</code> , <code>GetProfiles</code> and <code>GetStreamUri</code> (pointing to <code>rtsp://camsim:8554/c003</code>)
Vendor NVR API	REST channels 1 and 2 with <code>/status</code> and <code>/snapshot.jpg</code> (rendered with Pillow: perspective road, moving vehicles, OSD, blurred when the signal is weak) behind an API key; <code>/simulate?state=online weak offline</code> drives health demos
Public HLS	A public test stream stands in for an existing municipal VMS (C004)

24.2 Traffic

Producer	Behaviour
AI analytics engine	About 20 vehicles per minute (10 in production) over C001, C002, C003, C005, C006. Each vehicle is read 1–4 times within a couple of seconds (exercising dedup). 0.8 % are watchlisted plates, 30 % of those with an injected OCR error (exercising fuzzy matching). 4 % of cycles add a face-recognition event (10 % of them watchlisted persons); 5 % add a person or object detection with matching labels. Delivery is "mixed": half over REST with retries and stable event ids, half over the Redis Stream.
Edge agent	Heartbeats for C006 every 10 s with fps, bitrate, packet loss, uptime, temperature and firmware; can be paused
Dashcam (D001)	Drives a 12-junction patrol loop through Ahmedabad West, ticking every 3 s at 4× real time, speed varying 0–72 km/h. Each tick: a GPS heartbeat; a 45 % chance of an ANPR read of surrounding traffic at its current position (0.4 % watchlisted); ADAS rules — overspeed above 60 km/h in a 50 zone, harsh braking (3 %, a 25–35 km/h drop), collision warning (1.5 %), drowsiness (0.8 %)

24.3 Seed data

Four departments; three users; the `analytics-simulator` API client with all three scopes; cameras C001–C006 and D001 across Ahmedabad and Gandhinagar; 15 synthetic watchlist entries (12 vehicles using the reserved "XX" series, 3 persons with `FR-SYN` references); and a day of history (420 detections and four handled alerts) so a fresh install has meaningful charts. The seed is idempotent and never overwrites existing data.

25 Video gateway, proxy and Compose

25.1 Gateway configuration (infra/gateway/mediamtx.yml)

API on 9997, metrics on 9998, RTSP 8554, LL-HLS 8888 (always remux, 7×1 s segments, 200 ms parts), WebRTC 8889 with media on 8189 UDP/TCP (advertising `PUBLIC_IP` in production), playback server 9996, recording defaults (fMP4, 1 s parts, 60 s segments, 6 h), and the HTTP auth hook to the backend with only `metrics` excluded.

25.2 Edge proxy (infra/caddy/Caddyfile)

Route	Handling
<code>/api/*</code>	Internal paths and metrics answer 404; bodies ≤ 2 MB; proxied to the backend (WebSockets included)
<code>/media/*</code>	Prefix stripped, proxied to gateway LL-HLS; redirects rewritten back under <code>/media</code>
<code>/webrtc/*</code>	Bodies ≤ 64 KB; proxied to gateway WebRTC; redirects rewritten
<code>/playback/*</code>	Proxied to the gateway playback server
<code>everything else</code>	The React build with SPA fallback and the strict Content-Security-Policy

Headers on every response: HSTS (one year in production, zero on localhost), `nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: no-referrer`, `Cross-Origin-Opener-Policy`, a restrictive `Permissions-Policy`, and no `Server` header. TLS: Caddy's internal CA on localhost, automatic Let's Encrypt for a real domain.

25.3 Compose files and scripts

`docker-compose.yml` defines the stack (chapter 6). `docker-compose.prod.yml` overrides it for a public server: production mode, ports 80/443 and 8189, no backend port, one-year HSTS, public WebRTC address, gentler simulated traffic, higher login rate limit (all users arrive through a few proxy IPs on Vercel). Scripts: `gen-env.sh` (secrets), `camera.sh` (outage simulation), `deploy-vm.sh` (one-command server install), `backup.sh` (nightly dump), `configure-vercel.sh` (point the Vercel console at a backend).

PART VI

Quality, security, operations

How the platform is secured, tested, deployed, scaled and kept running — and what is left to do.

26 Security model

Defence in depth: each layer assumes the one in front of it may fail.

Threat	Defence
Password guessing	Per-IP rate limit, per-account lockout (5 failures → 15 min), bcrypt cost 12, constant-time path for unknown users
Stolen session token	8-hour expiry; server-side revocation on logout and password change; token kept in sessionStorage; strict CSP against script injection
Privilege escalation	Role check on every route; wrong token type rejected; API keys limited to explicit scopes
Camera credential leak	Fernet encryption at rest; never serialised; inline URL credentials refused; credentials injected only by the gateway
Unauthorised video access	Every HLS / WebRTC / recording request verified by the backend against a 5-minute, single-camera token
Reconfiguring streams	Gateway control API requires the backend's secret
SSRF via camera endpoints	Platform hosts, loopback, link-local (incl. cloud metadata), multicast and reserved addresses refused when saved and after DNS resolution before every request, including each redirect hop; snapshot proxy relays images only
Malicious XML	<code>defusedxml</code> for all ONVIF responses
Oversized / malformed input	Length limits on every field; request-body limits at the proxy; strict schemas; generic 500 responses
Clickjacking, sniffing, referrer leaks	Frame denial, nosniff, no-referrer (map tiles send origin only), COOP
Exposed services	Localhost-only binding by default; data tier unpublished; internal paths blocked; three segmented networks
Container breakout impact	Non-root users, all capabilities dropped, no-new-privileges, memory limits
Secrets in Git	<code>.env</code> ignored and generated randomly; staged files scanned against real secrets before commits
Vulnerable dependencies	Pinned versions; <code>pip-audit</code> and <code>npm audit</code> report 0 known vulnerabilities
Repudiation	Append-only audit log of every important action with actor and IP

Accepted risks: lockout can be abused to lock an account for 15 minutes (production would use SSO with MFA); map tiles reveal the viewed area to OpenStreetMap (self-host tiles for air-gapped use); environment variables are visible to host administrators (use a secrets manager in production).

27 Testing and CI

38 automated tests run against an in-memory configuration (SQLite, no Redis, gateway disabled) in about 10 seconds.

File	What it proves
test_auth_cameras.py	Bad passwords rejected; authentication required; onboarding hides credentials and is audited; inline credentials and wrong schemes refused; API-key bulk onboarding with per-item errors; viewers cannot modify; disabling is audited; playback tokens are camera-scoped and the media-auth hook enforces it
test_ingestion.py	Validation errors (unknown camera, confidence, naive and future timestamps); idempotent retries; repeat reads folded; watchlist match raises an alert with cool-down; fuzzy match lowers severity; low confidence does not alert; acknowledge / resolve workflow and role checks; trace order and impossible travel; batch and statistics; bulk resolve
test_dashcam.py	GPS heartbeats move the camera and build a track; events and alerts are geolocated where the van was; ADAS events stored; half a GPS fix rejected; clips refused for non-recordable cameras
test_security.py	Account logout; server-side logout; forged and wrong-type tokens; seven SSRF payloads; private camera networks still allowed; API-key scope validation; over-long passwords; security headers
test_retention.py	Old events pruned, alert evidence kept
test_matcher.py	Edit-distance-1 logic: substitution, insertion, deletion, trailing insertion, non-matches

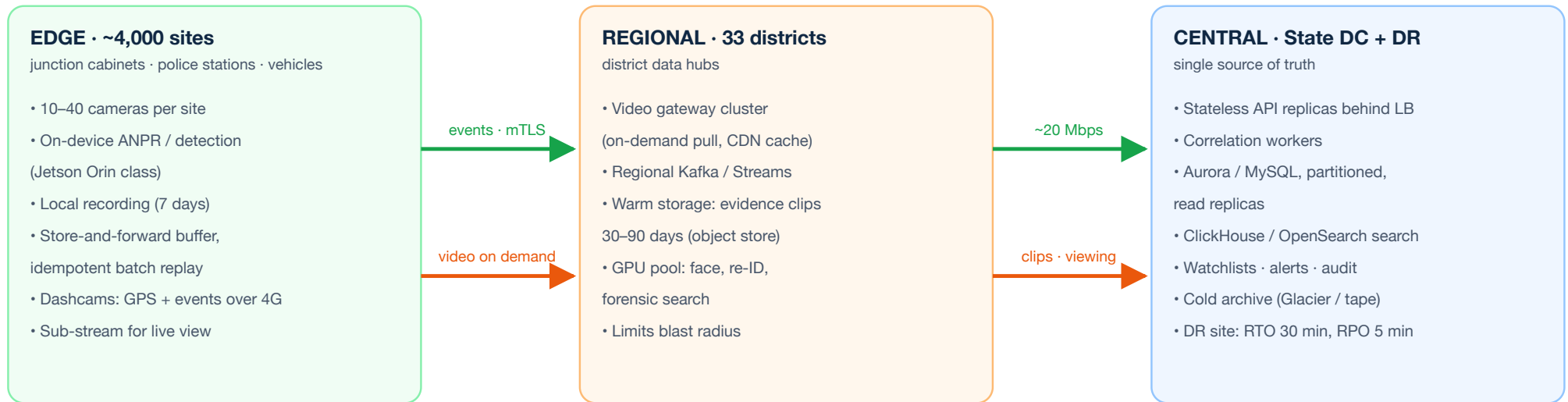
CI (GitHub Actions, on every push): backend lint (Ruff), format check and tests; simulator lint and compile; frontend ESLint and production build (TypeScript strict); and a build of every Docker image. **Beyond unit tests**, every change during development was verified against the running stack with a headless browser (pages, real-time updates, video playback, WebRTC, recordings, maps) and live API checks (token scoping, blocked routes).

28 Deployment

Option	How
Local	<code>make up</code> → <code>https://localhost:8443</code> . Trust Caddy's local root certificate once to avoid the browser warning.
Public link from this machine	Cloudflare quick tunnel (free, no account): <code>cloudflared tunnel --url https://localhost:8443 --no-tls-verify --http-host-header localhost --origin-server-name localhost</code> ; available while the machine is on; the address changes on restart.
Always-on server	Any Ubuntu VM (x86 or ARM; e.g. a free-tier cloud VM): <code>scripts/deploy-vm.sh <repo> <email></code> installs Docker, swap, firewall rules, automatic updates, sets <code>SITE_ADDRESS=<ip>.sslip.io</code> for a free real certificate, starts the production stack and schedules backups. Re-running it updates the server.
Vercel (console only)	<code>scripts/configure-vercel.sh <backend-host></code> generates <code>frontend/vercel.json</code> (HTTP rewrites, security headers) and <code>.env.vercel</code> (direct WebSocket URL, because Vercel cannot proxy WebSockets); import the repository with root directory <code>frontend</code> .

Operationally the stack restarts itself, is memory- and disk-bounded, renews certificates, and has health endpoints suitable for a free uptime monitor (`/api/ready`). See `docs/DEPLOYMENT.md` for step-by-step instructions.

29 Scaling to 80,000 cameras



The prototype already has these seams: adapters (edge / vendor integration), batch API (store-and-forward), Redis Streams consumer groups (→ Kafka), heartbeat-mode health, on-demand gateway pulls.

Figure 7 — Edge, regional and central tiers

29.1 The reasoning

80,000 cameras $\times$ 2 Mbps would be about 160 Gbps if all video went to one place. The design keeps video at the edge (recorded locally, viewed on demand through regional gateways) and moves only metadata: at an assumed 4,000 ANPR events per second (peak 12,000), the central uplink is about 20 Mbps and the database takes about 345 million rows a day. Every seam needed for this already exists in the prototype:

Prototype mechanism	At scale
Source adapters	Edge boxes and vendor SDKs plug in as adapters
Batch API + idempotency	Edge store-and-forward during WAN outages, safe replay
Redis Streams consumer group	Kafka partitioned by camera, many consumers
Stateless API + Redis fan-out	N replicas behind a load balancer, autoscaled
Leader-elected health monitor	Sharded by camera id per region
On-demand gateway pulls, snapshots	Bandwidth only where someone is watching; 2G/4G sites
Indexed MySQL, retention	Monthly partitions, read replicas, ClickHouse/OpenSearch for long-range search

GPU sizing, storage tiers, high availability, disaster recovery, security architecture and an indicative state-wide cost ($\approx$ ₹40 Cr one-time + ₹18–22 Cr per year, about ₹2,500 per camera per year) are detailed in `docs/SCALABILITY.md` and the architecture PDF.

30 Troubleshooting

Problems actually met during development, and their fixes.

Symptom	Cause	Fix
Browser: "Your connection is not private" with no Proceed button	Local certificate not trusted; an earlier one-year HSTS policy pinned localhost	Trust "Caddy Local Authority" in the keychain; clear localhost at <code>chrome://net-internals/#hsts</code> ; localhost now sends HSTS max-age=0
Map shows "Access blocked / 403"	OpenStreetMap requires a Referer; the site sends none	Fixed: tiles use <code>referrerPolicy=strict-origin</code>
<code>docker compose build</code> hangs on macOS	Docker Desktop's credential helper waits on a Keychain prompt	Approve the prompt, or build with a config that has no <code>credsStore</code>
Builds and tools extremely slow	Project folder on an iCloud-synced Desktop	Move the repository out of Desktop/Documents
Page goes black when an alert arrives	(Fixed) toast links rendered outside the router	Toaster inside the router + app-wide error boundary
Many open alerts	Simulated traffic with nobody acknowledging	Alerts → select all → Resolve with a note (bulk, audited)
C004 OFFLINE	Public test stream needs internet	Check connectivity or set <code>PUBLIC_HLS_URL</code>
WebRTC stays on "LL-HLS (WebRTC unavailable)"	UDP/TCP 8189 not reachable	Open the port; the player has already fallen back to LL-HLS
Port 8443 in use	Another local service	Set <code>HTTPS_PORT</code> in <code>.env</code>
"Account locked for 15 minutes"	Five wrong passwords	Wait, or reset with <code>python -m app.manage set-password</code>

31 Limitations and roadmap

Limitations of the prototype

- AI inference and camera feeds are simulated (the brief does not require model training); the interfaces are the real ones.
- ONVIF discovery probes configured hosts instead of multicast WS-Discovery.
- Recordings are a 6-hour hot tier; warm and cold tiers are designed, not built.
- Departments are modelled but data access is not yet restricted per department.
- The dashcam drives straight lines between junctions (no road snapping).

Roadmap

#	Next step	Value
1	Real edge inference (YOLO + OCR on NVIDIA DeepStream)	Live ANPR from real footage into the same API
2	Evidence clips to object storage with legal hold	Chain of custody beyond 6 hours
3	SSO (OIDC) with MFA; department-scoped access	Multi-agency tenancy, stronger identity
4	Kafka + ClickHouse; partitioned MySQL	State-scale throughput and long-range search
5	PTZ control over ONVIF; appearance-based re-identification	Operator control; tracking beyond plates
6	Playwright end-to-end tests; always-on hosting	Regression safety; permanent demo

32 Glossary

Term	Meaning
ANPR	Automatic Number-Plate Recognition: a camera plus model that reads vehicle registration plates
RTSP	Real-Time Streaming Protocol: how IP cameras deliver video; browsers cannot play it directly
ONVIF	Industry standard for discovering and controlling IP cameras over SOAP web services
HLS / LL-HLS	HTTP Live Streaming: video as small files over HTTPS; Low-Latency HLS splits segments into parts for ~1–2 s delay
WebRTC / WHEP	Real-time browser media (sub-second); WHEP is the simple HTTP handshake for receiving a WebRTC stream
MediaMTX	Open-source media server used as the video gateway (and for the simulated cameras)
NVR / VMS	Network Video Recorder / Video Management System — existing systems that hold cameras
ADAS	Advanced Driver-Assistance Systems events from a dashcam (overspeed, harsh braking, collision warning, drowsiness)
JWT	JSON Web Token: a signed token carrying the user's identity and role
RBAC	Role-Based Access Control (admin, operator, viewer)
Idempotency	Sending the same request twice has the same effect as once
Dedup window	Period (60 s) during which repeat reads of the same identity at the same camera are folded into one event
Cool-down	Period (5 min) during which a new match for an open alert updates it instead of raising another
Edit distance / SymSpell	Number of single-character edits between two strings / a fast method to find all strings within a small edit distance using deletions
Haversine	Formula for the distance between two points on a sphere from their latitudes and longitudes
Hysteresis	Requiring a condition to persist (two failed probes) before changing state, to avoid flapping
Leader election	Making sure only one replica runs a background job, using a short-lived Redis lock
Pub/Sub	Publish/subscribe messaging: every subscriber receives every published message
Redis Stream / consumer group / DLQ	Durable message log; a group of consumers sharing its messages with acknowledgements; dead-letter queue for messages that cannot be processed
SSRF	Server-Side Request Forgery: tricking a server into connecting to internal resources
CSP / HSTS	Content-Security-Policy: which sources a page may load / HTTP Strict Transport Security: forces HTTPS

Fernet	Authenticated symmetric encryption (AES-128-CBC + HMAC-SHA256) used for camera credentials
bcrypt	Deliberately slow password-hashing function
Alembic	Database migration tool for SQLAlchemy
sslip.io	Free DNS service where <code>1-2-3-4.sslip.io</code> resolves to 1.2.3.4, enabling a real HTTPS certificate for an IP address

End of handbook. Companion documents in the repository: `docs/Netra-Architecture.pdf` , `README.md` , `docs/API.md` , `docs/SECURITY.md` , `docs/SCALABILITY.md` , `docs/DEPLOYMENT.md` , `docs/DEMO_SCRIPT.md` .